



TOHMA BIOMASS POWER PLANT

Document Prepared by



EKI Energy Services Limited

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1 PROJECT DETAILS

1.1 Summary Description of the Project

TOHMA Biomass Power Plant project (hereafter referred to as the project activity) is a biogas-to-energy project that generates renewable energy (electricity) by using biogas obtained from anaerobic digestion of animal manure. Animal manure collected from multiple animal farm is treated in anaerobic digester and the biogas produced is utilized in cogeneration unit for thermal and electric energy generation. Whereas the generated electricity is fed to the national grid of Turkey, thermal energy is used internally for to meet up the process requirement (warming of the digester).

The project is being implemented by TOHMA Biyogas Enerji Limited Şirketi, as the Project Owner, in Yeşilyurt/Bindal district, Malatya province, in Turkey.

The common practice for the farm owners to manage the manure generated at the animal farms is storing the manure in anaerobic lagoons adjacent to the farms. The project, whose primary feedstock is manure generated at farms in Yeşilyurt, provides an environmentally friendly solution to this manure management problem. While providing sustainable development benefits to the host communities and the host country, the proposed project activity reduces greenhouse gas (GHG) emissions mainly by avoiding greenhouse gas (GHG) emissions and replacing the fossil fuel-dependent electricity.

The project activity is implemented in the borders of the central district of Malatya province, Yeşilyurt District, Bindal Neighborhood in Turkey.

, The manure produced in eighty numbers of neighboring livestock farms Of approximately 765 tons (cattle manure and poultry) are fed into the anaerobic digesters at the project site. Animal manure are collected in special sewage trucks equipped with close-tanks to prevent any odor and/or manure leakages and to be fed into the anaerobic digesters at the proposed biogas power plant. The project activity has three biogas engines installed at the biogas power plant, with 12,000 MWe.

The project began to generate electricity on June 29th, 2021, which is the date of commissioning of gas engines and is considered as the start date of the project activity. According to Electricity Generation License, the biogas generator system has the maximum operational capacity 12,000 MWe. The annual feed-in electricity is estimated to be 90,000 MWh.

The project has obtained the “Environmental Impact Assessment (EIA) not required” issued by the Ministry of Environment and Urbanization in Turkey on July 28th, 2022, and the Electricity Generation License (EGL) issued by the Ministry of Energy and Natural Resources in Turkey on June 10th, 2021. This project adopts the renewable crediting period of 7 years. The expected average annual emission reductions are 105,535 tCO₂eq/y. Accordingly, the project is expected

to generate 738,745 tCO₂eq emissions reduction during the first crediting period and 2,216,235 tCO₂eq emissions reduction throughout its three crediting periods for a total of 21 years.

1.2 Sectoral Scope and Project Type

Sectoral scope: 13 – Waste handling and disposal

Methodology: AM0073 “Approved baseline and monitoring methodology – GHG emission reductions through multi-site manure collection and treatment in a central plant”, Version 01¹. The project is not a grouped project.

1.3 Project Eligibility

The project is a biogas project, capturing Greenhouse Gases (GHGs), methane in particular, generated from animal manure in anaerobic conditions and utilizing them via cogeneration units, for electricity generation, which is eligible under Version 4.3 of the VCS Standard². According to methodology AM0073 Version 01³, the project is eligible since it receives wastes from almost 80 different farms in the city and all wastes are treated in a central treatment plant.

Since the project activity avoids methane and carbon-di-oxide, that are greenhouse gases identified under Kyoto Protocol the project activity is eligible under the scope of the Version 4.3 of VCS Standard⁴.

Additionally, the project is supported by methodology approved under an approved under CDM program. Therefore, the project is eligible under scope of VCS program.

1.4 Project Design

- ☒ The project includes a single location or installation only
- ☐ The project includes multiple locations or project activity instances, but is not being developed as a grouped project
- ☐ The project is a grouped project

Eligibility Criteria

The project is not a grouped project and hence not applicable

¹ <https://cdm.unfccc.int/methodologies/DB/2N19WQ6DCXNYRNJVZQQOHG7TK0Q2D8>

² https://verra.org/wp-content/uploads/2022/06/VCS-Standard_v4.3.pdf

³ <https://cdm.unfccc.int/methodologies/DB/2N19WQ6DCXNYRNJVZQQOHG7TK0Q2D8>

⁴ https://verra.org/wp-content/uploads/2022/06/VCS-Standard_v4.3.pdf

1.5 Project Proponent

Organization name	TOHMA Biyogaz Enerji Limited Şirketi
Contact person	Halil Söyler
Title	Manager
Address	Bindal mahallesi 1. Kümeevler no:68 Yeşilyurt/ MALATYA
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1.6 Other Entities Involved in the Project

Organization name	EKI Energy Services Limited
Contact person	Mr. Manish Dabkara
Title	Managing Director & Chief Executive Officer
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1.7 Ownership

Both the approval of the Environmental Impact Assessment (EIA) not required (EIA Exemption Certificate) issued by the Ministry of Environment and Urbanization in Turkey and the Electricity Generation License (EGL) issued by the the Ministry of Energy and Natural Resources in Turkey established the project ownership to TOHMA BİYOGAZ ENERJİ ÜRETİM ŞİRKETİ. Accordingly, these official documents establish the property and contractual rights both in the plant and equipment that generate GHG emission reductions.

1.8 Project Start Date

Project Start Date: 29/06/2021

The project activity at TOHMA Biogas project site was commissioned and started operation on 29-June-2021. Hence the project start date is defined as the earliest commissioning date within all the units.

1.9 Project Crediting Period

This project adopts the crediting period of 7 years, twice renewable for a total of 21 year. The first crediting period accordingly starts from 29/06/2021 and ends on 28/06/2028 (the start and end dates included).

1.10 Project Scale and Estimated GHG Emission Reductions or Removals

The estimated annual GHG emission reductions/removals of the project are:

- ☐ <20,000 tCO₂e/year
- ☐ 20,000 – 100,000 tCO₂e/year
- ☒ 100,001 – 1,000,000 tCO₂e/year
- ☐ >1,000,000 tCO₂e/year

Project Scale	
Project	
Large project	√

Year	Estimated GHG emission reductions or removals (tCO ₂ e)
29/06/2021-31/12/2021	53,780
2022	105,535
2023	105,535
2024	105,535
2025	105,535
2026	105,535
2027	105,535
01/01/2028-28/06/2028	51,756
Total estimated ERs	738,745
Total number of crediting years	7
Average annual ERs	105,535

1.11 Description of the Project Activity

Biogas is formed by a biological process. By being deprived of oxygen (anaerobe), a gas mixture called biogas emerges from the organic mass. This process, which is common in nature, also takes place in swamps, seabed, dung pits, and rumen of ruminants. Meanwhile, the organic mass is almost completely converted to biogas by several microorganisms. In addition, certain amounts of energy (heat) and new biomass are formed. Biogas consists of methane (50-75% Vol.) and carbon dioxide (25-50% Vol.). In addition, biogas also contains small amounts of hydrogen, hydrogen sulphide, ammonia, and traces of other gases. The composition is mainly determined by the materials used, the fermentation process and different technical applications. The biogas formation process takes place in many stages. In the meantime, for the whole process to develop in a way that does not cause any negativity, the individual degradation stages must be very compatible with each other.

As described in section 1.1, the scenario existing prior the project's implementation is the continuum of common practice for the livestock farm owners in the region to use manure generated at their farms to land application after storing the manure in anaerobic lagoon resulting in greenhouse gases generated from animal manure, methane especially, were released directly into atmosphere. Emissions associated with the baseline scenario are the CH₄ emissions due to the atmospheric release of methane sourced from animal manure left to decay anaerobically and CO₂ emissions due to the power generation by the national grid, which fossil fuel-burning power plants dominate.

Through the project's implementation, animal manure collected from farms are transported through covered trucks to prevent any odor and/or manure leakages and is received in waste collection area. The methane previously released from animal manure left to decay anaerobically will be captured via anaerobic digestion and utilized to produce thermal and electric energy through a cogeneration system in the biogas power plant. Captured methane via digestion process will be sent to desulfurization unit to be purified before electricity production by gas engines. In addition, the electricity generated at the plant will be directly fed to the national grid, avoiding and/replacing the dispatch of the same amount of electricity from fossil-fuel-based power plants connected to the national grid. Moreover, waste heat produced in the cogeneration unit will warm up anaerobic digesters. By-products of the plant, which are solid and liquid fertilizers from the manure waste stored in anaerobic digesters, will be used in agricultural activities within neighboring farms to increase the quality of the soil. Thus, none of the equipment installed within the project boundary exists in the baseline scenario.

The project will consist of the installation of units given Table 1 below.

Table 1 Number of Units in Plant

Number of Units	Unit Name
1	Wheel washing unit
1	Weighbridge
3	Waste receiving tanks
2	Equalization tanks
1	Grit chamber
4	Fermenter tank
1	Hygiene tank
4	Seperator
1	Liquid fermented product storage unit (Lagoon)
1	Solid fermented product storage unit
2	Electric generator system (Cogeneration)
1	Desulphurization system
1	Flare

- **Wheel Washing Unit**

A wheel washing unit is established in order to ensure the disinfection of vehicles entering and leaving the facility. The wheel washing unit consists of a pool washing system. Washing wastewater to be collected in the settling pond connected to the wheel washing unit is drawn by a vacuum truck and sent to the nearest wastewater treatment plant.

- **Weighbridge**

Animal wastes are brought to the facility by closed box trucks belonging to the company or to be rented, and vehicles will enter the waste reception area after they are weighed at the weighbridge at the entrance. All wastes brought to the facility will be accepted after they are weighed and recorded. A data recording system is established that includes information such as the source, amount, and access method of the waste accepted to the facility. The weighbridge to be used in the facility has 60 tons capacity.

The amount and type of waste coming to the facility every day is recorded in the waste acceptance unit. For this reason, the design was made with a waste storage capacity of approximately 3 days, taking into account any malfunctions and maintenance situations. The waste reception unit, which is designed in reinforced concrete and impermeable, will be covered with manholes and will have a biofilter in which the foul-smelling air is filtered.

- **Waste Receiving Tanks**

Wastes shall be accepted to the facility in a way that does not pose an environmental risk and necessary control systems shall be established to monitor the processing of wastes. The wastes to be accepted to the facility will be taken to the waste areas to be made within the facility according to the type of waste after the waste acceptance process. Waste with high liquid content such as Cow Manure will be taken directly to equalization tanks. Wastes with high solid content such as Cattle Manure, Laying Chicken Manure, Broiler Manure will be accepted in waste reception area 1. Plant wastes such as Fruit Pickle, Corn Silage, Sugar Beet, Grass Silage, which will be accepted to the facility, will be accepted into waste reception area 2.

Waste reception area-1 (30m x 20m) has a total capacity of 1800 m³ with a height of 3 meters. The total amount of waste that will come to this area daily is approximately 484,376 m³ (333,33+104,16 +46,88) m³. It has the capacity to store the wastes planned to be stored in this area for approximately 3.5 days.

Waste reception area- 2 (15m x 10 m) has a total capacity of 450 m³ with a height of 3 meters. The total amount of waste that will come to this area daily is 24.49 m³ (5.92+6.73+5.92+5.92) m³.

It has the capacity to store the wastes planned to be stored in this area for approximately 18 days. At the same time, the waste reception area will be taken to 3 for liquid wastes. Waste reception area 3 has the capacity to store at least 1 day's waste.

- **Equalization Tanks**

Equalization tanks have a volume of 603 m³. The equalization tanks, where the waste will be collected after the reception area, will be in a closed, mostly underground and impermeable C₃₀ reinforced concrete structure and their interiors will be covered with epoxy sealing material. After each waste is taken to the landfill, it will be prepared according to the appropriate recipe and sent to the equalization tanks before entering the feed according to the type and quality of the waste (vegetable wastes will be subjected to shredding process).

As the system as a whole is interconnected, the different feeding and flow regime can be determined by the operator according to the type of waste fed.

- **Grit Chamber**

The wastes homogenized in the equalization tanks will be fed to the digesters after passing through the grit chamber unit to separate the marble dust that comes with the chicken manure. The volume of the grit chamber unit will be 50 m³. After the wastes are kept in the grit chamber unit for 1 hour, the separated sand will be sent to the residual waste area.

- **Fermenter Tank**

The anaerobic digestion system in the facility will consist of 4 vertical type reactors, all of which will be digesters. 4 digester reactors will be used to process the waste and convert the organic load in the structure into biogas by breaking down after a biological process. Anaerobic digestion process is carried out in the Anaerobic Digestion System Unit. Anaerobic digestion system consists of external heating system, anaerobic digestion reactors and mixers, internal and external biological biogas desulphurization systems, biogas storage and transfer system, flare stages. Each of the reactors planned to be installed at the facility will have an inner diameter of 32 meters, an area of 804.25 m², and will be built in a cylindrical shape. Anaerobic digesters consist of four vertical, stainless, central agitator reactors that will serve as the main digester reactors and designed to circulate with cycle pumps from the bottom to dredge the entire bottom.

- **Liquid Fermented Product Storage Lagoon**

The fermented product can be used as a liquid or a solid by dehydrating. After separation, liquid fermented product is sent to the lagoon to store and used for agricultural use if it meets the necessary conditions. Otherwise, it will be sent to the wastewater treatment plant. The storage unit has a volume of 70,000 m³ and is 10 m deep with impermeable layer.

- **Solid Fermented Product Storage**

The solid fermented product storage area will have a total of 1200 m² storage area with an area of 30 m x 40 m. With 4 meters high walls, it will have a volume of 4800 m³ and a storage capacity of 51 days.

- **Hygiene Tank**

Fermented Product Hygiene will be exposed to at least 1 hour at 70°C. After hygiene, 729 tons/day of the liquid phase (889 tons/day) will be sent to the open process water aeration tanks and will be fed back to the system after ammonia removal is achieved.

- **Separator Unit**

Within the scope of the project, 4 separators with a capacity of 35 m³/h will be used.

- **Desulphurization system**

The biogas stored on the digesters at 3-5 millibars in a 2-stage inflatable roof in a PVC membrane structure with a very low methane permeability (400 cm³ /m²/bar/24) is purified from H₂S by internal desulphurization method after being sent to the external double-layered gas balloon by means of pumps. Afterwards, after the rough water is taken from the condensation shaft over the gas line, it is sent to the gas cooling and filtration unit for precise dehumidification. Here, a blower system is used both to pressurize the gas on the line and to send it to the gas engines with a positive pressure of 80-120 millibars. In addition, the gas pressurized in the blower is measured volumetrically by passing through a flow meter before being sent to the engines, and the methane and H₂S content is continuously monitored by a gas analyzer. The gas lines are laid underground to maintain a constant condensation temperature up to the blower room in the building and to prevent the condensate from freezing and clogging the lines.

These pipes, which are made of high-pressure resistant PE material, will be selected from stainless steel material as they go from the blower chamber to the top. The condensed water collected in the condensation shafts on the gas lines will be transferred back to the system with the existing submersible pumps. Since H₂S is a very reactive gas, it is important to remove it as soon as it is formed. Therefore, there is internal desulfurization in the chosen process and a very small amount of air (1-2% by volume) is transferred to all membranes thanks to a blower system. Sulfur bacteria found in inflatable roofs convert H₂S from gaseous form to collapsible elemental form and rapidly separate it from the gas. In this way, the H₂S level is reduced below 200 ppm, which is the limit for most gas engines. In addition, this will increase the quality of the product, as sulfur passes into the fermented product in solid form. The biogas leaving the system will be included in the existing biogas system again after the chemical scrubber, and its moisture will be removed in the dehumidifier. Since the return water flow rate is lower in the bio-trickling filter, the humidity in the biogas will also be less.

- **Cogeneration Unit**

A cogeneration system with an installed power of 3.17 MWe will be used to convert the produced gas into electricity and waste heat. The gas engines to be used will have a total efficiency of more than 85% and will have a flue gas silencer system and an anti-vibration system. The gas engine that will meet the desired standards in terms of NO_x emission values will be selected in a way that will provide the highest level of waste heat recovery.

There will be 2 gas (cogeneration) engines in the facility. The gas coming out of the reactors will be sent to the engine building by fixing a certain pressure by means of a blower. Gas (cogeneration) engines produce electricity and heat by burning biogas. The low voltage (400V) electricity produced will be amplified by booster transformers and given to the medium voltage line (34500V) and sold to the state.

1.12 Project Location

The project was implemented over an area of 40,000 m² square meter of within acceptance boundary of TOHMA Biogas Project in Malatya Province, Yeşilyurt District, Bindal Neighborhood in turkey.

Table 1. Coordinates of TOHMA Biogas Power Plant Project

Point Number	Latitude	Longitude
1	38° 29'54.4232"	38° 12'23.8983"
2	38° 29'54.1583"	38° 12'22.0110"
3	38° 29'46.4051"	38° 12'15.3712"
4	38° 29'48.4124"	38° 12'11.9432"

5	38° 29' 50.2188"	38° 12' 11.3452"
6	38° 29' 59.0606"	38° 12' 07.2886"
7	38° 30' 01.2433"	38° 12' 06.2571"
8	38° 30' 07.3680"	38° 12' 15.6906"
9	38° 30' 00.9047"	38° 12' 17.6175"
10	38° 29' 58.8358"	38° 12' 21.4242"
11	38° 29' 56.3691"	38° 12' 23.1989"



Figure 1 Satellite Image of Project Area



Figure 2. Location of Tohma Biogas Power Plant project

1.13 Conditions Prior to Project Initiation

The scenario existing prior to the start of the implementation of the project activity is anerobic decomposition of animal manure. Before the proposed project activity, livestock farm used to store the produced manure in anaerobic lagoon near by the farm. Therefore, greenhouse gases produced from anerobic decomposition of animal manure were released directly into the atmosphere.

In addition, the amount of electricity generated using biogas technology would have been generated by power plants connected to the Turkish public power grid, which is heavily dependent on fossil fuel resources.

The baseline scenario is the same as the scenario existing prior to the start of the implementation of the project activity. Please refer to Section 3.4 (Baseline Scenario) for details.

1.14 Compliance with Laws, Statutes and Other Regulatory Frameworks

The two of main regulatory approvals for such project types to prove that they are in compliance with relevant laws, statutes and regulatory frameworks in Turkey are the Environmental Impact Assessment and the Electricity Generation License whose obtainments are required for projects being in compliance with other relevant legal requirements.

Environmental Impact Assessment (EIA) not required: July 28th ,2022

The Electricity Generation License (EGL): June 10th, 2021.

1.15 Participation under Other GHG Programs

1.15.1 Projects Registered (or seeking registration) under Other GHG Program(s)

The project has not been registered or is seeking registration under any other GHG programs.

1.15.2 Projects Rejected by Other GHG Programs

The project has not been rejected by any other GHG programs.

1.16 Other Forms of Credit

1.16.1 Emissions Trading Programs and Other Binding Limits

The project does not reduce GHG emissions from activities that are included in an emissions trading program or any other mechanism that includes GHG allowance trading.

1.16.2 Other Forms of Environmental Credit

The project has not sought or received another form of GHG-related credit or renewable energy certificates.

1.17 Sustainable Development Contributions

1.17.1 Sustainable Development Contributions Activity Description

The project contributes to three of the UN Sustainable Development Goals (SDGs) in total. These SDGs are given below;

- SDG 7 Affordable & Clean Energy,
- SDG 8 Decent Work & Economic Growth,
- SDG 13 Climate Action,

The project generates 90,000 MWh/year which is directly fed into the city grid. The generated electrical energy is directly fed to the Turkish national grid (SDG 7.2.1.).

The project activity has created some permanent jobs for maintenance and operation during the monitoring period. In addition, the employees were provided with occupational health and safety trainings (SDG 8.0).

The project is expected to generate 105,535 metric tons of CO₂eq in total per annum (SDG 13.0).

The project developer is going to complete the VERRA's "Sustainable Development Contributions Report"²⁶ for the project. Hence, the more detailed data on project's SDG Contribution can be found both in the contribution report and the final monitoring report once they are submitted.

1.17.2 Sustainable Development Contributions Activity Monitoring

It is defined in section 17.1.

Table 2: Sustainable Development Contributions

Row number	SDG Target	SDG Indicator	Net Impact on SDG Indicator	Current Project Contributions	Contributions Over Project Lifetime
<i>Sequential row number</i>	<i>SDG Target number</i>	<i>Number and text of SDG indicator or, if no official SDG indicator is applicable, user-defined indicator</i>	<i>Indicate the project's contribution to the SDG Indicator (implemented activities to increase or decrease)</i>	<i>Brief description of the quantifiable impact of the project's activities related to the SDG indicator, during the monitoring period.</i>	<i>Brief description of the cumulative quantifiable impact of the project's activities related to the SDG indicator, over the project lifetime.</i>
2)	7.2	7.2.1 Renewable energy share in the total final energy consumption	The project being a renewable based power generation unit feeding power to the grid enhances the renewable energy share in the final energy consumption	Supply of 113,178 MWh of renewable electricity to the grid	Cumulative supply of 113,178 MWh of renewable electricity to the grid.
3)	8.5	8.5.2 Unemployment rate, by sex, age and persons with Disabilities	The project activity results in temporary and permanent employment opportunity during the construction and operation phase	Creating employment opportunities for 10 persons	Cumulative employment opportunities for 10 persons created.

4)	8.6	8.6.1 Proportion of youth (aged 15-24 years) not in education, employment, or training	The employment opportunities created through the project results in employment of youths	Creating employment opportunities for 2 youth	Cumulative employment opportunities for two youths created
5)	13.2	13.2.2 Total greenhouse gas emissions per year	Implemented activities avoided atmospheric release of methane through its capture and utilization for power generation. The generated electricity using renewable energy resources avoids greenhouse gas emission by refraining generation of electricity in fossil fuel-based power plant connected to the grid	avoidance of 132,714 tCO2e for the monitoring period	Cumulative emission avoidance of 132,714 tCO2e for monitoring period of 29/06/2021-30/09/2022

1.18 Additional Information Relevant to the Project

Leakage Management

The potential leakage sources of the project activity include the manure transportation of manure to the facility, manure storage, leakage from the anaerobic digesters, and land application of the treated residues.

Animal manure collected from the supplier farms is transported to the project site via confined trucks, and the collected manure is utilized within less than 24 hours in the plant. Additionally, the anaerobic digesters are confined units that are equipped with heat and leakage insulation. Thus, this leakage has been excluded in the emission reductions calculation as per the applied methodology and tool (please refer to Section 4.3). Therefore, a leakage management plan and implementation of leakage and risk mitigation measures in terms of land application of the treated residues are unnecessary.

Commercially Sensitive Information

No commercially sensitive information has been excluded from the public version of the project description.

Further Information

Not Applicable

2 SAFEGUARDS

2.1 No Net Harm

Ministry of Environment and Urbanization in Turkey decided that “EIA not required” on July 28th, 2022. In accordance with the Turkish laws and regulations, EIA not required shall be only made if a project subjected to the approval does not make any negative environmental and socio-economic impacts. A review was made regarding the project and the precautions to be taken against environmental impacts in the project introduction file were deemed sufficient. It has been determined that it is not necessary to prepare an EIA report, and it has been decided that "EIA not required" for the project. Considering the fact that the proposed project activity already obtained EIA not required by the Ministry of Environment and Urbanization in Turkey, it is therefore possible to claim that there is no net harm linked to the project.

2.2 Local Stakeholder Consultation

The Project Owner decided to organize a local stakeholder consultation meeting to inform the stakeholders about technical and non-technical issues about the facility. The meeting date was

determined as 12.04.2022; the project owner prepared invitation letters and invited all stakeholders within the project. To reach out to the locals living in the Yeşilyurt district and the Malatya province, the project personnel publicly announced the LSC meeting in the company website. Personnel made the public announcement, consisting of the LSC meeting's agenda, location, and date, along with a brief project description on 3rd March, 2022. During the 8th of March, announcements were carried out to reach the identified stakeholders. First, invitation letters were posted in the mosque and mukhtar, which are the most frequently visited spots in the Malatya region, to inform the local people. Also, invitation letters were hand-delivered to village mukhtars. During the hanging of the invitation letters, all citizens found in the places were told by the project officer in order to register and participate in the meeting. Meeting invitations sent by e-mails to the relevant institutions, organizations, non-governmental organizations, and individuals. Due to the constant communication of the project authorities, national authorities, regional and provincial administrations were personally called and invited.

Minutes of the LSC Meeting

The local stakeholder information meeting was held on 12th April 2022 as agreed. At the beginning of the meeting, it was explained in detail to the participants meeting, how questions about the project would be collected during the meeting and how they could give feedback. The plant manager presented the project details and improvements of the project site. He provided detailed information on how the TOHMA Biogas Plant functions and talked about the technical information that the audience could easily understand. After his presentation, Climate & Sustainability Consultant Ahmet Hamdi Zembil provided insights into the historical backgrounds of Sustainable Development and Climate Change at the international arena, including establishing UNFCCC in 1994 and the 1997 Kyoto Protocol. Then, he briefly explained what the UN's Sustainable Development Goals were. During his presentation, he informed audiences about how climate change occurred and the result we are facing now. He continued by explaining the main greenhouse gas types and consequences of these gases on the world and gave an end to her speech by providing insight into stopping climate change by increasing renewable energy potential, reforestation, and energy efficiency. After her presentation, Mr. Zembil continued to start his speech later by explaining to the audience that one of the central subjects and aims in organizing the meeting was to assess the Project's contribution in achieving sustainable development goals and tackling the climate crisis. He gave information about carbon offsetting mechanism and carbon crediting with the clarifications of how sustainable development contribution is essential in host country/communities' and 'additionality.' While doing so, to simplify these complex terms to an appropriate level that met the stakeholders' understanding, he/she*** sketchily explained what VERRA/VCS was, how it functioned, and the main requirements for projects like TOHMA Biogas Plant to access its certification.

Name-Surname	Gender	Job Title
Ahmet Huzeyfe Karakoç	Male	Environmental Engineer
Mehmet Kılıç	Male	Mukhtar of Sütluçe Neighborhood
Abubekir Çetin	Male	Farmer

Osman Kılınç	Male	Farmer
Nihat Kılınç	Male	Farmer
Mustafa Kılınç	Male	Worker
Mehmet Çetin	Male	Farmer
Ramazan Kılıç	Male	Farmer
İsmail Çetin	Male	Farmer
Halit Kılınç	Male	Farmer
İlyas Tamer	Male	Farmer
Davut Alkan	Male	Mukhtar
Ahmet H. Zembil	Male	Environmental Engineer
Ayşe Yılmaz	Female	Housewife
Sibel Geçici	Female	Housewife
Fadime Aydoğan	Female	Housewife
Türkan Gözel	Female	Cook
Pınar Cetiz	Female	Housewife
Şerafettin Geçici	Male	Retired

2.3 Environmental Impact

The environmental matters in Turkey are jurisdiction of the Ministry of Environment and Urbanization. The Law defines among others the general requirements of the Environmental Impact Assessment (EIA) states specific requirements according to the type of project, location, size and other characteristics.

Ministry of Environment and Urbanization in Turkey decided that “EIA not required” on July 28th, 2022. In accordance with the Turkish laws and regulations, EIA not required shall be only made if a project subjected to the approval does not make any negative environmental and socio-economic impacts. A review was made regarding the project and the precautions to be taken against environmental impacts in the project introduction file were deemed sufficient. It has been determined that it is not necessary to prepare an EIA report, and it has been decided that "EIA not required" for the project. Considering the fact that the proposed project activity already obtained EIA not required by the Ministry of Environment and Urbanization in Turkey, it is therefore possible to claim that there is no net harm linked to the project.

2.4 Public Comments

To be defined.

2.5 AFOLU-Specific Safeguards

Since the project is a non-AFOLU project, this section is not required.

3 APPLICATION OF METHODOLOGY

3.1 Title and Reference of Methodology

Applied approved baseline and monitoring methodologies: AM0073 “Approved baseline and monitoring methodology – GHG emission reductions through multi-site manure collection and treatment in a central plant”, Version 01⁵

Sectoral scopes: 13 – Waste handling and disposal

Applied tools:

- “Project emissions from flaring”, Version 04.0⁶;
- “Tool to calculate the emission factor for an electricity system”, Version 07.0⁷;
- “Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation”, Version 03.0⁸;
- “Tool for the demonstration and assessment of additionality”, Version 07.0.0⁹;
- “Methodological tool: Investment analysis”, Version 11.0¹⁰;
- “Methodological tool: Common practice”, Version 03.1¹¹;

3.2 Applicability of Methodology

The methodology is applicable under the following conditions:

Applicability of AM0073 – v01.0			
Ref.	Application Criteria	Applicability	Justification
1.	Farms where livestock populations, comprising of cattle, buffalo, swine, sheep, goats, and/or poultry, are managed under confined conditions;	Applicable	Project area has livestock population in confined condition. Hence applicable

⁵ <https://cdm.unfccc.int/methodologies/DB/2N19WQ6DCXNYRNJVZQQOHG7TK0Q2D8>

⁶ <https://cdm.unfccc.int/methodologies/DB/2N19WQ6DCXNYRNJVZQQOHG7TK0Q2D8>

⁷ <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-06-v4.0.pdf>

⁸ <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-05-v3.0.pdf>

⁹ <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-01-v7.0.0.pdf>

¹⁰ <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-27-v11.0.pdf>

¹¹ <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-24-v1.pdf>

Applicability of AM0073 – v01.0			
Ref.	Application Criteria	Applicability	Justification
2.	Farms where manure is not discharged into natural water resources (e.g. rivers or estuaries);	Applicable	Manure is left for anaerobic decaying not discharged into natural water resources. Hence applicable
3.	Farms where animal residues are treated under anaerobic conditions;	Applicable	Manure is left for anaerobic decaying in anaerobic lagoon in the baseline scenario. Hence applicable
4.	The annual average temperature in the site where the anaerobic manure treatment facility in the baseline existed is higher than 5 °C;	Applicable	The annual average temperature of baseline site where manure is collected from multiple farms in Malatya is 13.7 °C ¹² . Hence, the proposed project activity complies with this criteria/condition
5.	In the cases where the baseline anaerobic treatment system is an open lagoon, the lagoon depth shall be greater than 1 m;	Applicable	Baseline anaerobic treatment system is open lagoon and depth is greater than 1m, same has been explained in detail in Baseline section. Hence applicable.
6.	The retention time of the organic matter in the baseline anaerobic treatment systems should be at least 30 days;	Applicable	Baseline anaerobic treatment system has retention time 30 days. Hence applicable.

¹² <https://www.mgm.gov.tr/veridegerlendirme/il-ve-ilceler-istatistik.aspx?m=MALATYA>

Applicability of AM0073 – v01.0			
Ref.	Application Criteria	Applicability	Justification
7.	If residues are stored in between collection activities, storage tanks shall comprise outdoor open equipments;	N/A	No treated residue in baseline. Hence condition not applicable.
8.	If the treated residue is used as fertilizer in the baseline, project proponents must ensure that this end use remains the same throughout the project activity;	Not Applicable	The animal manure is stored in anerobic lagoon therefore the criteria is not applicable.
9.	Sludge produced during the project activity shall be stabilized through thermal drying or composting, prior to its final disposition/application;	Applicable	Sludge produced during the project activity is stabilized through thermal drying, prior to its final disposition/application. Hence applicable
10.	The AWMS/process in the project case should ensure that no leakage of manure waste into ground water takes place, e.g., the lagoon should have a nonpermeable layer at the lagoon bottom;	Applicable	The AWMS/process in the project case does not have leakage of manure waste into ground water takes place. Hence applicable
11.	CERs shall be claimed by the Central Treatment Plant managing person/entity, only. Other parties involved must sign a legally binding declaration that they will not claim CERs from the improved animal waste treatment practices. Such declarations shall be verified by the DOE during the validation, and these documents shall be valid throughout the whole crediting period.	Applicable	A declaration has been submitted by Central Treatment Plant managing person/entity to the DOE. Also, a manure supply agreement with farm owners will be submitted to the DOE which states that owner of central treatment plant has the rights of VCUs.

Applicability of AM0073 – v01.0			
Ref.	Application Criteria	Applicability	Justification
12.	Anaerobic manure treatment systems without methane recovery in the farms are the most plausible baseline scenario	Applicable	Anaerobic manure treatment systems without methane recovery are the most plausible baseline scenario. Please refer section 3.4. Hence applicable
13.	Finally, this methodology is only applicable if the application of the procedure to identify the baseline scenario results in that anaerobic manure treatment systems without methane recovery in the farms are the most plausible baseline scenario	Applicable	Anaerobic manure treatment systems without methane recovery are the most plausible baseline scenario. Hence applicable

Apart from the methodology, following tools are applicable for the project activity:

- Project emissions from flaring – v04.0

Ref.	Application Criteria	Applicability	Justification
1.	Methane is the component with the highest concentration in the flammable residual gas; and	Applicable	Methane component is highest in the project case
2.	The source of the residual gas is coal mine methane or a gas from a biogenic source (e.g. biogas, landfill gas or wastewater treatment gas).	Applicable	In this gas source of gas is biogas
3.	The tool is not applicable to the use of auxiliary fuels and therefore the residual gas must have sufficient flammable gas present to sustain combustion. In the case of an enclosed flare, there shall be operating specifications provided by the manufacturer of the flare and these shall be followed by the project participant.	Applicable	In the project, there is no usage of auxiliary fuel

- Tool to calculate project or leakage CO₂ emissions from fossil fuel combustion – v03.0

Ref.	Application Criteria	Applicability	Justification
1.	This tool provides procedures to calculate project and/or leakage CO ₂ emissions from the combustion of fossil fuels. It can be used in cases where CO ₂ emissions from fossil fuel combustion are calculated based on the quantity of fuel combusted and its properties. Methodologies using this tool should specify to which combustion process j this tool is being applied.	Applicable	Project is displacing the electricity from the fossil fuel dominated grid.

- Tool to calculate the emission factor for an electricity system – v07.0

Ref.	Application Criteria	Applicability	Justification
1.	This tool may be applied to estimate the OM, BM and/or CM when calculating baseline emissions for a project activity that substitutes grid electricity, i.e. where a project activity supplies electricity to a grid or a project activity that results in savings of electricity that would have been provided by the grid (e.g. demand-side energy efficiency projects).	Applicable	The project activity substitutes grid electricity by supplying renewable power to grid. Hence this criterion is applicable.
2.	In case of CDM projects the tool is not applicable if the project electricity system is located partially or totally in an Annex I country.	Applicable	Project is not a CDM project. Hence, it is applicable.

- Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation – v03.0

Ref.	Application Criteria	Applicability	Justification
1.	If emissions are calculated for electricity consumption, the tool is only applicable if one out of the following three scenarios applies to the sources of electricity consumption: (a) Scenario A: Electricity consumption from the grid. The electricity is purchased from the grid only, and either no captive power plant(s) is/are installed at the site of electricity consumption or, if any captive power plant exists on site, it is either not operating or it is not physically able to provide electricity to the electricity consumer; (b) Scenario B: Electricity consumption from (an) off-grid fossil fuel fired captive power plant(s). One or more fossil fuel fired captive power plants are installed at the site of the electricity consumer and supply the consumer with electricity. The captive power plant(s) is/are not connected to the electricity grid; or (c) Scenario C: Electricity consumption from the grid and (a) fossil fuel fired captive power plant(s). One or more fossil fuel fired captive power plants operate at the site of the electricity consumer. The captive power plant(s) can provide electricity to the electricity consumer. The captive power plant(s) is/are also connected to the electricity grid. Hence, the electricity cons	Applicable	Scenario A, Electricity consumption from the grid fits for the project activity
2.	This tool can be referred to in methodologies to provide procedures to monitor amount of electricity generated in the project scenario, only if one out of	Applicable	Scenario I, Electricity is supplied to the grid fits for the project activity

Ref.	Application Criteria	Applicability	Justification
	the following three project scenarios applies to the recipient of the electricity generated: (a) Scenario I: Electricity is supplied to the grid; (b) Scenario II: Electricity is supplied to consumers/electricity consuming facilities; or (c) Scenario III: Electricity is supplied to the grid and consumers/electricity consuming facilities		
3.	This tool is not applicable in cases where captive renewable power generation technologies are installed to provide electricity in the project activity, in the baseline scenario or to sources of leakage. The tool only accounts for CO2 emissions.	Not Applicable	This does not fit for the project case since it exports the electricity to the grid.

3.3 Project Boundary

As noted above, the baseline and monitoring methodology applied to the proposed project activity is AM0073 “GHG emission reductions through multi-site manure collection and treatment in a central plant”, version 01.0. In accordance with the Methodology, the spatial extent of the project boundary encompasses:

- The central treatment plant;
- The livestock farms where the waste is generated and disposed in the baseline;
- The site of the biogas combustion or energy generation facility;
- The manure storage tanks;
- The road itineraries between the manure collection points and the central treatment plant;

The relevant GHG sources included in or excluded from the project boundary are shown on the Table below.

Source		Gas	Included?	Justification/Explanation
Baseline	Direct emissions from the manure treatment processes	CO ₂	No	CO ₂ emissions from the decomposition of organic waste are not accounted.
		CH ₄	Yes	The major source of emissions in the baseline
		N ₂ O	No	Direct and indirect N ₂ O emissions are not accounted
		Other	-	-
	Emissions from electricity consumption /generation	CO ₂	Yes	Electricity may be consumed from the grid or generated onsite in the baseline scenario
		CH ₄	No	Excluded for simplification. This is conservative.
		N ₂ O	No	Excluded for simplification. This is conservative.
		Other	-	-
	Emissions from thermal energy generation	CO ₂	No	Excluded for simplification. Although thermal energy will be generated through the implementation of the proposed project activity, no emission will be claimed for this, as noted in this PD. Hence, this exclusion shall be considered as conservative.
		CH ₄	No	
		N ₂ O	No	
		Other	-	
Project	Emissions from on-site electricity use	CO ₂	Yes	These emissions are accounted since the proposed project may consume electricity from the national grid.
		CH ₄	No	Excluded for simplification. This emission source is assumed to be very small.
		N ₂ O	No	Excluded for simplification. This emission source is assumed to be very small.
		Other	-	-
	Emissions from thermal energy generation	CO ₂	No	Excluded for simplification. As noted above, the project will also generate thermal power. Although the proposed project activity will not claim any emission
		CH ₄	No	
		N ₂ O	No	

Source		Gas	Included?	Justification/Explanation
		Other	-	reduction from the thermal energy generation, it will not consume thermal energy from any outside source. The thermal energy generated within the boundaries of the proposed project activity will be used for the project's auxiliary heat consumption. This exclusion shall be considered as conservative.
	Emissions from the waste treatment processes	CO ₂	No	CO ₂ emissions from the decomposition of organic waste are not accounted.
		CH ₄	Yes	The emissions from anaerobic digesters and aerobic treatment are accounted.
		N ₂ O	Yes	Direct and indirect N ₂ O emissions are accounted.
		Other	-	-
	Emissions from manure / waste residue transportation	CO ₂	Yes	May be an important emission source. Hence, they are accounted.
		CH ₄	No	Excluded for simplification. This emission source is assumed to be very small.
		N ₂ O	No	Excluded for simplification. This emission source is assumed to be very small.
		Other	-	-
	Emissions from residue from anaerobic digester composting	CO ₂	No	CO ₂ emissions from the decomposition of organic waste are not accounted.
		CH ₄	No	The proposed project activity does not involve composting.
		N ₂ O	No	The proposed project activity does not involve composting.
		Other	-	-
	Emissions from manure storage tanks	CO ₂	No	CO ₂ emissions from the decomposition of organic waste are not accounted.
		CH ₄	No	Since manure is not stored more than 24 hours these emissions are not accounted. This emission source is assumed to be very small.
		N ₂ O	No	Excluded for simplification. This emission source is assumed to be very small.
		Other	-	-

In addition, the diagram of the project boundary below shows clearly the physical locations of the various installations or management activities taking place as part of the project activity.

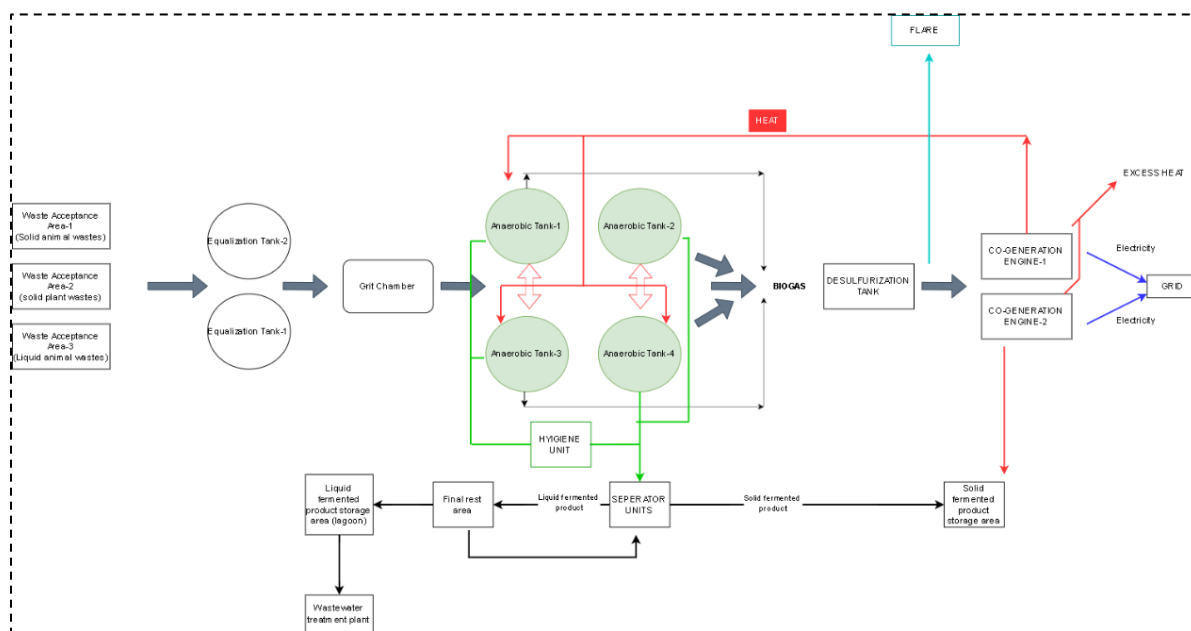


Figure 3. Flow Chart of the Plant

3.4 Baseline Scenario

The proposed project activity has two major GHG emission reduction components. The first one is the reduction of methane emissions resulting from improved animal manure management, and the second one is the use of biogas recovered from improved animal manure management for generating power and feeding the electricity generated directly to the national electricity grid network. AM0073 “Approved baseline and monitoring methodology – GHG emission reductions through multi-site manure collection and treatment in a central plant”, Version 01.0, states that “Baseline scenario should be identified from the perspective of the owner of central treatment plant, as well as from the perspective of the multiple livestock farms owners”. Project participants shall apply the following steps to identify the baseline scenario:

Step 1: Identify plausible alternative scenarios

As per Methodologies of AM0073, Version 01.0, the following alternatives or combinations of these alternatives shall, inter alia, be considered for determining baseline scenario for managing the manure:

Description of the Alternative Scenarios	Plausible and credible alternative?	Justification
For the Owner of Central Treatment Plant		

The proposed project activity not being registered as a VERRA/VCS activity	Yes	This is a plausible scenario.
If applicable, continuation of the previous situation (no project activity or alternatives undertaken)	Yes	This is a plausible scenario.
For the Owners of the Livestock Farms		
The proposed project activity not being registered as a VERRA/VCS activity	Yes	This is a plausible scenario.
All other plausible and credible alternatives to the project activity scenario, including the common practices in the relevant sector. In doing so, the complete set of possible manure management systems listed in the 2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories (Chapter 10, Table 10.17)¹³ should be considered. In drawing up a list of possible scenarios, possible combinations of different Animal Waste Management Systems (AWMS) should be considered. The list includes the following scenarios, excluding Uncovered Anaerobic Lagoon (previous situation, hence, it is not accounted: • Pasture/Range/Paddock • Deep Bedding • Composting • Poultry manure with litter • Poultry manure without litter • Aerobic Treatment • Liquid/Slurry • Pit Storage under animal confinements • Solid Storage • Dry lot • Burned for fuel	Yes	Explanations for each manure management systems are given below.

¹³ The 2006 IPCC Guidelines for National Greenhouse Gas Inventories is refined by IPCC. Accordingly, Chapter 10, Table 10.17 of the 2006 IPCC Guidelines for National Greenhouse Gas Inventories is updated with Chapter 10, Volume 4, Table 10.14 of the 2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories.

- Daily spread - Anaerobic Digestion – Biogas (Project Activity)		
If applicable, continuation of the previous situation	Yes	It is a plausible and credible alternative since the owner of the livestock farms would continue to store manure generated at their farms in uncovered anaerobic lagoons, as further explained below.

For the owner of the central treatment plant:

The plausible alternative scenarios for the owner of the central treatment plant are identified as no project activity undertaken and the proposed project activity not being registered as a VERRA/VCS activity, which are in compliance with all mandatory applicable legal and regulatory requirements.

For the owner of the livestock farms:

Uncovered anaerobic lagoon –Animal waste manure management prior to the project activity

The Regulation on Livestock Farms¹⁴ implemented in 2006 in Turkey vaguely requests livestock farms to have a lagoon to collect the manure generated at the farms. While the regulation did not provide any technical specifications for such lagoons, it was stated that the lagoons' interior walls which stayed underground shall be of concrete foundation -in order to prevent the manure from leaking into soil and underground water resources. Thus, it is justifiable to state that there is currently no legal regulation for the livestock farms that defines an anaerobic lagoon design. Although it varies based on the scale of the farm, the overall depth of such lagoons at the farms around the project site is approximately 2 meters. Since there is no legal regulation to mandate the livestock farm owners to capture and/or utilize methane generated at these lagoons, continuing to use anaerobic lagoons as manure management -without making any modification to their existing manure management systems, is viable and reasonable for animal farm owners.

Both liquid and solid manure generated at cattle and poultry farms are stored in the lagoons for 2-3 months till they are evacuated – usually to agricultural fields. This system complies with

¹⁴ <https://www.resmigazete.gov.tr/eskiler/2006/08/20060809-4.htm>

environmental and legal regulations in Turkey and therefore applicable as a baseline and/or alternative scenario for the proposed project activity.

Pasture/Range/Paddock – Excluded due to regulatory requirements

The regulations in Turkey state that the animal manure should be collected in concrete made lagoons/ponds, this type of manure management system is not in compliance with the the Regulation on Livestock Farms.

Deep Bedding – Excluded due to regulatory requirements

This manure management system is not allowed due to the Regulation on Livestock Farms that states manure shall be kept only on concrete made lagoons / ponds.

Composting – Excluded since it can only be applied as a secondary / continuous stage

Because the composting system can only be applied as a subsequent stage after manure storage and requires expertise, this technology is not applicable for the baseline scenario of this project.

Poultry manure with litter – Excluded due to regulatory requirements

This manure management system is not allowed due to the Regulation on Livestock Farms that states manure shall be kept only on concrete made lagoons / ponds.

Poultry manure without litter – Excluded due to regulatory requirements

This manure management system is not allowed due to the Regulation on Livestock Farms that states manure shall be kept only on concrete made lagoons / ponds.

Aerobic Treatment – Excluded due to high energy requirements and lack of sufficient land

Aerobic treatment is not a preferred manure management in Türkiye since forced aeration requires high electricity consumption and natural aeration requires aerobic and facultative ponds and wetland systems which are limited in Türkiye.

Liquid/Slurry – Excluded due to regulatory requirements

This manure management system is not allowed due to the Regulation on Livestock Farms that states manure shall be kept only on concrete made lagoons / ponds.

Pit storage under animal confinements – Excluded due to regulatory requirements

This form of manure management was used to be applied in animal farms. However, in line with the Regulation on Livestock Farms made on livestock farms, this form of manure management system is not allowed anymore. Those who had an installed system of this kind had given legal notice to reform their systems accordingly.

Solid storage – Excluded due to regulatory requirements

This manure management system is prohibited by the Regulation on Livestock Farms¹⁵ which states that manure shall be stored on concrete made structures to prevent any harm on the environment and human health.

Dry lot – Excluded due to regulatory requirements

Because the Regulation on Livestock Farms in Turkey enforces the use of concrete storage areas, dry lot manure management system, which is a paved or unpaved open confinement area, is not a viable option for the livestock farm owners.

Burned for fuel – Excluded due to regulatory requirements

This manure management system is not in line with current regulations. Livestock farm owners are prohibited to use this manure management method because of the emission of hazardous gasses emitted from burning dung cakes.

Daily Spread – Excluded due to regulatory requirements

According to the Regulation on Livestock Farms, this form of manure management system is not allowed since untreated animal manure contains high nitrogen level and harms the organic quality of soil as well as underground aquifers.

The plausible alternative scenarios for the owner of the livestock farms are identified **as the continuation of using the uncovered anaerobic lagoons-based manure management system and the proposed project activity not being registered as a VERRA/VCS activity**, which are in compliance with all mandatory applicable legal and regulatory requirements.

Step 2: Barrier Analysis

As per Methodologies of AM0073, Version 01.0, and the Methodological Tool: “Combined tool to identify the baseline scenario and demonstrate additionality”, version 07.0¹⁶ barriers that would prevent the implementation of alternative scenarios should be identified and alternative scenarios prevented by these barriers should be eliminated.

Sub-step 2.a. Identify barriers that would prevent the implementation of alternative scenarios

The following barriers have been identified for the plausible alternative scenarios:

Technological barriers:

The continuation of the prior manure management system, uncovered anaerobic lagoons with an average retention time of 2-3 months, requires very low technological knowledge and expertise. On the other hand, implementation of the proposed project activity requires sufficient infrastructure and maintenance expertise. Furthermore, the proposed project activity poses risk

¹⁵ <https://www.resmigazete.gov.tr/eskiler/2006/08/20060809-4.htm>

¹⁶ <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-02-v7.0.pdf>

of technological failure as the anaerobic digester systems include controlling and constant maintenance of anaerobic bacteria growth. Any imbalance in the process might lead to shut down of the facility. Because of such barriers the project activity is not applicable for the baseline scenario for the owner of the livestock farms. Thus, farm owners continue to use uncovered anaerobic lagoons, as their manure management system.

Sub-step 2.b. Eliminate alternative scenarios which are prevented by the identified barriers

As a result of the barrier analysis, the remaining alternative scenarios are:

- The continuation of the previous situation (uncovered anaerobic lagoons-based manure management system) and the proposed project activity not being registered as a VERRA/VCS activity for the owner of the central treatment plant.
- The continuation of the previous situation (uncovered anaerobic lagoons-based manure management system) for the farm owners.

According to the Methodological Tool: “Combined tool to identify the baseline scenario and demonstrate additionality”, version 07.0¹⁷, if more than one alternative scenario is not prevented by any barrier and the project activity is not the first-of-its-kind and the remaining scenarios include the proposed project activity undertaken without being registered as a Verra/VCS project activity, Step 3 Investment Analysis must be applied. Thus, the investment analysis is conducted below.

Step 3: Investment Analysis

The details of the investment analysis are given in Step 2 of Section 3.5 of the PD. The result of investment analysis clearly demonstrates that without the generation of revenues from the sale of Verified Carbon Units (VCUs), the project activity is not financially attractive

Step 4: Baseline revision at renewal of crediting period

As per the methodology of AM0073 “Approved baseline and monitoring methodology – GHG emission reductions through multi-site manure collection and treatment in a central plant”, Version 01.0¹⁸, “At the renewal of each crediting period, the relevance of baseline scenario identified above will be assessed taking into account change in the relevant national and/or sectoral regulations between two crediting periods as well as any increase in the animal stock above the pre-project animal stock” (pg. 6).

This project adopts the renewable crediting period of 7 years, twice renewable for a total of 21 years.

¹⁷ <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-02-v7.0.pdf>

¹⁸ <https://cdm.unfccc.int/methodologies/DB/2N19WQ6DCXNYRNJVZQQOHG7TKOQ2D8>

The procedures and analyses made in accordance with the applied methodologies and tools documented in this section results that anaerobic manure treatment systems in the farms whilst no implementation of the central plant is the most plausible baseline scenario.

3.5 Additionality

Section 3.13 in VCS Standard v4.3. states that “A project activity is additional if it can be demonstrated that the activity results in emission reductions or removals that are in excess of what would be achieved under a ‘business as usual’ scenario and the activity would not have occurred in the absence of the incentive provided by the carbon markets” (2022:34). Moreover, Section 3.13.1 clearly mandates that “Additionality shall be demonstrated and assessed in accordance with the requirements set out in the methodology applied to the project” (ibid.)

In the case of the proposed project activity, the Methodology of AM0073 “Approved baseline and monitoring methodology – GHG emission reductions through multi-site manure collection and treatment in a central plant”, Version 01.0, is applied to the proposed project activity. Demonstration of additionality is accordingly carried as per the methodological Tool: “Tool for the demonstration and assessment of additionality”, Version 07.0¹⁹.

The following steps from the Tool are completed below:

STEP 0 – Demonstration whether the proposed project activity is the First-of-its-kind;

STEP 1 – Identification of alternative scenarios;

STEP 2 – Investment analysis;

STEP 3 – Barrier analysis;

STEP 4 – Common practice analysis.

STEP 0 – Demonstration whether the proposed project activity is the First-of-its-kind

This step is skipped since the project is not first-of-its-kind.

STEP 1 – Identification of alternative scenarios

Regarding the identification of alternative scenarios, Step 1 of Section 3.4 of the PD has already demonstrated the identification step. As the result of the alternative scenario analysis, the continuation of the previous manure management system is found to be the plausible alternative scenario to the project activity.

STEP 2 – Investment analysis

Investment analysis is carried as per the methodological Tool: “Tool for the demonstration and assessment of additionality”, Version 07.0.

Sub-step 2a: Determine appropriate analysis method

¹⁹ <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-01-v7.0.0.pdf>

There are three options which can be applied for the investment analysis. These are as follow:

- Option I. Simple Cost Analysis;
- Option II. Investment Comparison Analysis;
- Option III. Benchmark Analysis

Since the Project generates economic benefits from sales of electricity, the simple cost analysis is not applicable. Also, since the baseline scenario involves the use of electricity from the national grid as usual, no alternative investment is considered. Therefore, investment comparison analysis is not suitable.

Consequently, the Option III, namely benchmark analysis is decided as the most suitable investment analysis method for this project. In this respect, the financial indicator (Equity IRR in this case) of the proposed project will be compared with the benchmark value.

Sub-step 2b: Option III. Apply Benchmark Analysis

While applying the Benchmark Analysis, Option III, the Equity IRR is selected as the financial indicator for the demonstration of the additionality of the project as permitted in the additionality tool.

For benchmark analysis, the after-tax equity IRR for biogas projects of 20% in the World Bank document²⁰ dated June 19th, 2017 (p40, table 3.3 of the referred document) has been used as the benchmark IRR.

The IRR inputs used are:

Table 3 List of IRR Inputs

Input name	Input Value	Reference Document
Energy production	24,960 GWh	Company information
Operational life time	48 years 11 months 11 days	Licence agreement ²¹ and temporary gas motor provisional acceptance document
Electricity sales price	13.3 USD cents/KWh for the first 10 years and 4.5 USD cents/KWh for the rest of the project life	Renewable Energy law no: 5346
O&M expenses	USD 1,699,377 annually	Company information
Tax rate	25%	Corporate tax law no: 5520
Total investment	USD 2,920,401	Company information
Electromechanical Equipment	USD 2,365,850	
Buildings	USD 554,551	

²⁰ <https://documents1.worldbank.org/curated/en/799701498842988254/pdf/ICR00004069-06192017.pdf>

²¹ Licence agreement starts from 10 June 2021 and ends on 10 June 2070. The operation starts on 29 June 2021. Hence the operational lifetime is 48 years 11 months 11 days till the end of licence..

The company has a guaranteed tariff rate of 13.3 USD cents/kWh for the first 10 years of operations. After 10 years, we have used a tariff rate of 4.5 USD cents/kwh as our analyst estimate based on the market overview. The investments are financed by 100% equity. There is no debt financing. We have assumed that after every 8 years, there is a gas motor reinvestment and after 40 years, there is a reinvestment of constructional parts.

Sub-step 2c - Calculation and comparison of financial indicators (only applicable to options II and III)

Based on the parameters in the table of list of IRR inputs, the equity after-tax IRRs is shown below:

	IRR
Without carbon financing	5.66%

The IRR of the total investment of the proposed project without carbon financing revenue is 5.66%, lower than the benchmark IRR of 20%, which means that the proposed project activity indeed faces significant financial barriers. Therefore, the project activity is not the most economically or financially attractive choice of investment.

Sub-step 2d - Sensitivity Analysis

The sensitivity analysis shall show whether the conclusion regarding the financial attractiveness is robust under reasonable variations in the critical assumptions. For the project four basic parameters have been selected as sensitive factors to check out the financial attractiveness:

- The electricity production
- Electricity selling price
- Annual operational costs
- Investment costs

Applying variations on investment costs, the range was considered to be in a bandwidth of +10% and -10% as advised in the CDM Methodological tool Investment analysis Version 11.0. The tables below show the variation of IRR when the four parameters fluctuate from -10% to +10%.

Table 4 The sensitivity analysis

	IRR Sensitivities				
Change in %	-10.0%	-5.0%	0.0%	5.0%	10.0%
Annual operational costs	2.98%	4.30%	5.66%	7.12%	8.76%
Electricity production	9.73%	7.37%	5.66%	4.27%	3.08%
Electricity selling price	9.73%	7.37%	5.66%	4.27%	3.08%
Investment costs	5.48%	5.57%	5.66%	5.75%	5.84%

Sensitivity analysis has been carried out for the stated costs and assumptions. It shows that under none of those scenarios the IRR of the project exceeds the Benchmark

Outcome of Step 2: Hence, the above figures have demonstrated that the project activity is not the most economically or financially attractive alternative.

STEP 3 – Barrier analysis

As per Paragraph 44 of the methodological Tool: “Tool for the demonstration and assessment of additionality”, Version 07.0, since the sensitivity analysis is concluded that the proposed project activity is unlikely to be financially/economically attractive, the PP has proceeded to Step 4.

STEP 4 – Common Practice analysis

Common practice analysis for the proposed project activity is conducted as per the methodological tool: “Common practice”, Version 03.1²². This methodological tool provides a stepwise approach for the analysis of the extent to which a proposed project type (e.g., technology or practice) has already diffused in the relevant sector and region (pg. 2).

Step 4.1: Calculate applicable capacity or output range as +/-50% of the total design capacity or output of the proposed project activity

The design capacity of the proposed project activity is 12,000 MWe. Accordingly, the applicable output range is from 6,000 MWe to 18,000 MWe.

Step 4.2: Identify similar projects (both CDM and non-CDM) which fulfil all the following conditions:

- a. The projects are in the applicable geographical area;
- b. The projects apply the same measures as the proposed project activity;
- c. The projects use the same energy source/fuel and feedstock as the proposed project activity, if a technology switch measure is implemented by the proposed project activity;
- d. The plants in which the projects are implemented produce goods and services with comparable quality, properties, and application areas (e.g., clinker) as the proposed project plant;
- e. The capacity or output of the projects is within the applicable capacity or output range calculated in Step 4.1.;
- f. The projects started commercial operation before the project design document (CDM-PDD) is published for global stakeholder consultation or before the start date of proposed project activity, whichever is earlier for the proposed project activity.

According to the official report of “Final List on Renewable Energy Plants, 2022²³”, published by Republic of Turkey Energy Market Regulatory Authority (EPDK), there are only 3 operational biogas/biomass plants whose installed capacities are within the proposed project activity’s applicable capacity range calculated in Step 4.1. (6,000 MWe to 12,000 MWe) in 2022 in Black

²² <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-24-v1.pdf>

²³ [https://www.epdk.gov.tr/Detay/DownloadDocument?id=4ETVklVlw00=\(Final List of YEKDEM-2022\)](https://www.epdk.gov.tr/Detay/DownloadDocument?id=4ETVklVlw00=(Final List of YEKDEM-2022))

Sea, East Anatolian and South East Anatolian Region where the regions near to project activity..
The list of the plants is given in the Table below:

No.	Name of the Plant	Total Installed Capacity (MWe)	Voluntary Emission Reductions Certification Scheme(s) Registry ^{24,25} (ID No.)
1	Ahi Suluova Biyogaz Tesisleri	6,637	GS5882
2	ÇORUM BES	6,020	N/A
3	Bingöl Biyogaz Tesisi	6,402	N/A

Step 4.3: within the projects identified in Step 2, identify those that are neither registered CDM project activities, project activities submitted for registration, nor project activities undergoing validation. Note their number, N_{all} .

Within the projects identified in the Table above, the projects which are neither registered CDM project activities, project activities submitted for registration, nor project activities undergoing validation are as follow:

1. Çorum BES
2. Bingöl Biyogaz Tesisi

Accordingly, it is noted that

$$N_{all} = 2$$

Step 4.4: within similar projects identified in Step 3, identify those that apply technologies that are different to the technology applied in the proposed project activity. Note their number N_{diff} .

Amongst the two projects identified in Step 4.3 above, those that apply technologies that are different to the technology applied in the proposed project activity, which is anaerobic digestion with the main feedstocks of animal waste, is as follow:

No project applies a different technology to the project activity. Thus,

$$N_{diff}=0$$

Step 4.5: calculate factor $F=1-N_{diff}/N_{all}$ representing the share of similar projects (penetration rate of the measure/technology) using a measure/technology similar to the measure/technology used in the proposed project activity that deliver the same output or capacity as the proposed project activity.

²⁴ Search via Gold Standard Project Registry; accessed on 20.09.2022 [GSF Registry \(goldstandard.org\)](https://goldstandard.org)

²⁵ Search via Gold Standard Project Registry; accessed on 120.09.2022 [Verra Search Page](https://verra.org)

In Step 4.3., N_{all} is noted as 2

In Step 4.4., N_{diff} is noted as 0

Accordingly, factor $F=1-N_{diff}/N_{all}$ is calculated as follows:

$$F = 1 - 0$$

$$F = 1$$

Whilst F is equivalent to 1, $N_{all}-N_{diff}$ is equivalent to 0.

Outcome of Step 4

The paragraph 18 of the methodological tool “Common practice”, Version 03.1²⁶ states that “the proposed project activity is a ‘common practice’ within a sector in the applicable geographical area if the factor F is greater than 0.2 and $N_{all}-N_{diff}$ is greater than 3” (pg. 5).

For the proposed project activity, $F=1$ is greater than 0.2, whilst $N_{all}-N_{diff}=0$ which is not greater than 3. Hence, the proposed project activity shall be considered as not a common practice in the region.

Hence, in accordance with the methodological tool “Tool for the demonstration and assessment of additionality”, Version 07.0²⁷, and the methodology of AM0073 “Approved baseline and monitoring methodology – GHG emission reductions through multi-site manure collection and treatment in a central plant”, Version 01.0²⁸, the proposed project activity shall be considered additional.

3.6 Methodology Deviations

No methodology deviation is applied in the project.

4 ESTIMATED GHG EMISSION REDUCTIONS AND REMOVALS

4.1 Baseline Emissions

The baseline emissions are:

- Emissions from cattle and chicken manure left to decay in anaerobic conditions at the farms;

²⁶ <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-24-v1.pdf>

²⁷ <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-01-v7.0.0.pdf>

²⁸ <https://cdm.unfccc.int/methodologies/DB/2N19WQ6DCXNYRNJVZQQOHG7TKOQ2D8>

- Emissions resulting from electricity generation.

The calculation of the baseline emissions for the proposed project activities are based on the methodology of AM0073 “Approved baseline and monitoring methodology – GHG emission reductions through multi-site manure collection and treatment in a central plant”, version 01.0.

4.1.a. Baseline emissions

In accordance with AM0073 “GHG emission reductions through multi-site manure collection and treatment in a central plant”, version 01.0, baseline emissions are calculated as the sum of CH₄ and N₂O emissions that would occur in the baseline animal waste treatment system and CO₂ emissions arising from heat and consumption.

According to the methodology AM0073, baseline emissions are calculated by using equation (1) and include the following sources:

Equation (1)

$$BE_y = BE_{AW,y} + BE_{\frac{elec}{heat}}^y$$

Where:

BE_y = Total baseline emissions in year y, in tCO₂e/year

$BE_{AW,y}$ = Baseline emissions attributable to animal waste treatment in year y, in tCO₂e/year

$BE_{\frac{elec}{heat}}^y$ = Baseline CO₂ emissions from electricity and/or heat generated/consumed in the baseline, in tCO₂e/year

4.1.a.1. Emissions from animal waste treatment

As per AM0073 “GHG emission reductions through multi-site manure collection and treatment a central plant”, version 01.0, the baseline is the AWMS identified through the baseline selection procedure.

Equation (2)

$$BE_{AW,y} = BE_{AW,CH_4,y} + BE_{AW,N_2O,y}$$

Where:

$BE_{AW,y}$ = Baseline emissions attributable to animal waste treatment in year y, in tCO₂e/year

$BE_{AW,CH_4,y}$ = Baseline methane emissions attributable to animal waste treatment in year y, in tCO₂e/year

$BE_{AW,N_2O,y}$ = Baseline N₂O emissions attributable to animal waste treatment in year y, in tCO₂e/year.

4.1.a.1.i Methane emissions from animal waste treatment

In accordance with AM0073 “GHG emission reductions through multi-site manure collection and treatment in a central plant”, version 01.0, manure management system in the baseline could be based on different treatment systems and on one or more stages. Therefore:

Equation (3)

$$BE_{AW,CH_4,y} = GWP_{CH_4} \times \rho_{CH_4} \times \sum_{j,LT} (MCF_j \times B_{0,LT} \times N_{LT,y} \times VS_{LT,y} \times MS\%_{BL,j})$$

Where:

$BE_{AW,CH_4,y}$	= Annual baseline methane emissions in tCO ₂ e/y
GWP_{CH_4}	= Global Warming Potential (GWP) of CH ₄
$\rho_{CH_4,n}$	= CH ₄ density (6.7x10 ⁻⁴ t/m ³ at room temperature (200C) and 1 atm pressure)
MCF_j	= Annual methane conversion factor (MCF) for the baseline AWMSj from 2019 Refinement to 2006 IPCC Guidelines for National Greenhouse Gas Inventories (IPCC 2019), Volume 4, Chapter 10, Table 10.17 (Updated), page 67 ²⁹
$B_{0,LT}$	= Maximum methane producing potential of the volatile solid generated, in m ³ CH ₄ /kg-VS-dm, by animal type LT
$N_{LT,y}$	= Annual average number of animals of type LT for the year y, expressed in numbers
$VS_{LT,y}$	= Annual volatile solid for livestock LT entering all AWMS [on a dry matter weight basis (kg-VS-dm/animal/year)], as estimated below
$MS\%_{BL,j}$	= Fraction of manure handled in AWMS type j in the baseline scenario

Estimation of $VS_{LT,y}$, $B_{0,LT}$ and MCF_j :

As per AM0073 “GHG emission reductions through multi-site manure collection and treatment in a central plant”, version 01.0, there are five ways to determine the $VS_{LT,y}$ value including the use of published data, estimation based on dietary intake of livestock, scaling default IPCC values to adjust for the site-specific average animal weight, utilizing default values of IPCC, and direct measurement. The project participants have chosen to use the 5th option. The reasoning behind the elimination of the first four option is as follows:

- There are no published country specific data for Turkey.
- Data related to dietary intake of livestock is not available. This is due to the fact that TOHMA Biomass Power Plant animal manure from multiple animal farms which cannot track or supply specific dietary intake data. Furthermore, farm owners tend to constantly change the livestock feed due to changes in its prices and availability in the Turkish market.
- The 3rd option is not applicable because it is not practical and/or doable to receive regular W_{site} data from the multiple farms that supply animal manure to the biogas

²⁹ 4 MCFj value was calculated based on the table 10.17 (Updated) from 2019 Refinement to 2006 IPCC Guidelines for National Greenhouse Gas Inventories (IPCC 2019), Volume 4, Chapter 10. The average temperatures were obtained from the Turkish State Meteorological Service <https://mgm.gov.tr/veridegerlendirme/il-ve-ilceler-istatistik.aspx?m=MALATYA>

project. While laws and regulations³⁰ mandate the farm owners to track the number of animals through ID numbers specific to each animal, there are no rules to track the weight of the animals. This is why the PPs are not able to request such information from the livestock farms as per the ways in which such data is requested to be obtained in the applied Meth.

- The 4th requires the use of IPCC default values for developed countries. The conditions for using the developed country values include;
 - The genetic source of the production operations livestock originates from an Annex I Party;
 - The farm use formulated feed rations (FFR) which are optimized for the various animal(s), stage of growth, category, weight gain/productivity and/or genetics;
 - The use of FFR can be validated (through on-farm record keeping, feed supplier, etc.);
 - The project specific animal weights are more similar to developed country IPCC default values.

However, it is not possible to ensure the genetic source and the use of FFR for each farm since the source of animal and used feed can vary depending on availability and feed prices. Because it is not possible to demonstrate that the project activity meets the conditions for using the developed country values, the 4th option is eliminated.

Consequently, direct measurement of $VS_{LT,y}$ is used as shown in Equation (4) below:

Equation (4)

$$VS_{LT,y} = W_{manure,LT} \times VS_{manure,LT} \times nd_y$$

Where:

$VS_{LT,y}$	= Annual volatile solid excretions on a dry-matter weight basis (kg-VS-dm/animal/yr)
$W_{manure,LT}$	= Average manure weight excreted by a defined population at the project site in kg/animal/day)
$VS_{manure,LT}$	= Average VS in the manure excreted by a defined population at the project site in kg-VS-dm per kg of manure; (calculated as per Annex 2 of the approved methodology of AM0073)
nd_y	= Number of days the central treatment plant was operational in year y

According to AM0073 “GHG emission reductions through multi-site manure collection and treatment in a central plant”, version 01.0, Maximum Methane Production Potential ($B_{0,LT}$) can be determined in two different ways including the use of default values from the IPCC Guidelines and direct measurement. The project participants choose to determine in the way that default

³⁰ [https://www.resmigazete.gov.tr/eskiler/2011/12/20111202-9.htm#:~:text=MADDE%2021%20%E2%80%93%20\(1\)%20Hayvan,g%C3%BCn%20i%C3%A7inde%20ilgili%20%C4%B0l%2F%C4%B0l%C3%A7e](https://www.resmigazete.gov.tr/eskiler/2011/12/20111202-9.htm#:~:text=MADDE%2021%20%E2%80%93%20(1)%20Hayvan,g%C3%BCn%20i%C3%A7inde%20ilgili%20%C4%B0l%2F%C4%B0l%C3%A7e)

values are used, and they are obtained from 2019 IPCC Refinement, Table 10.16, (p. 10.72, Other Regions, High Productivity Systems -specific to the country where the proposed project activity is implemented). The direct measurement method is eliminated due to a lack of data.

The values for Methane conversion factors (MCF_i) are obtained from 2019 IPCC Refinement, Table 10.17, (page 10.74), depending on the annual average temperature where the anaerobic manure treatment facility in the baseline is located. As noted in AM0073 “GHG emission reductions through multi-site manure collection and treatment in a central plant”, version 01.0, future revisions to the IPCC Guidelines for National Greenhouse Gas Inventories are considered. Note that a conservativeness factor is also applied by multiplying MCF values (estimated as per above) with a value of 0.94, to account for the 20% uncertainty in the MCF values as reported by IPCC 2006.

Determination of Annual Average number of animals (N_{LT}) is done following the equation below. The applied AM0073 methodology states that the annual average number of animals should be determined using the Equation 22 or 23 in the Methodology. Equation 23 can be applied “if the project developer can monitor in a reliable and traceable way the daily stock of animals in the farm, discounting dead animals and animals discarded from the productive process from the daily stock”. Because the project developer is not able monitor the number of animals in a reliable and traceable way, Equation 22 in the Methodology is used as demonstrated below.

Equation (5)

$$N_{LT} = N_{da} * \frac{N_p}{365}$$

Where:

N_{LT} = Annual average number of animals of type LT for the year y, expressed in numbers
 N_{da} = Numbers of days animal is alive in the farm in the year y, expressed in numbers
 N_p = Number of animals produced annually of type LT for the year y, expressed in numbers

Accordingly,

BE_{AW,CH4,y} is calculated as 50,569 tCO₂e/yr

4.1.a.1.ii N₂O emissions from manure management

Equation (6)

$$BE_{AW,N2O,y} = GWP_{N2O} * CF_{N2O-N,N} * 10^{-3} * (E_{N2O,D,y} + E_{N2O,ID,y})$$

Equation (7)

$$E_{N_2O,D,y} = \sum_{j,LT} (EF_{N_2O,D,j} * NEX_{LT,y} * N_{LT,y} * MS\%_{BI,j})$$

Equation (8)

$$E_{N_2O,ID,y} = \sum_{j,LT} ((EF_{4,j} + EF_{5,j}) * NEX_{LT,y} * F_{gasm} * MS\%_{BI,j})$$

Where:

$BE_{AW,N_2O,y}$	= Annual baseline N ₂ O emissions in tCO ₂ e/yr
GWP_{N_2O}	= Global Warming Potential (GWP) for N ₂ O
$CF_{N_2O-N,N}$	= Conversion factor N ₂ O-N to N ₂ O (44/28)
$E_{N_2O,D,y}$	= Direct N ₂ O emissions in kg N ₂ O-N/year
$E_{N_2O,ID,y}$	= Indirect N ₂ O emissions in kg N ₂ O-N/year
$EF_{N_2O,D,j}$	= Direct N ₂ O emission factor for the treatment system j of the manure management system in kg N ₂ O-N/kg N (estimated with site-specific, regional, or national data if such data is available, otherwise use default EF ₃ from IPCC 2019 Guidelines for National Greenhouse Gas Inventories , Chapter 10, Volume 4, Table 10.21-Updated) (as per the IPCC 2019 Guideline, EF ₃ value was considered as zero for uncovered anaerobic lagoon).
$NEX_{LT,y}$	= The annual average nitrogen excretion per head of a defined livestock population in kgN/animal/year estimated as described in Equation 10.30 from 2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas inventories (page 10.80).
$N_{LT,y}$	= Number of animals of type LT for the year y, expressed in numbers
$MS\%_{BI,j}$	= Fraction of manure handled in system j
$\% F_{gasm}$	= Percent of managed manure nitrogen for livestock category that volatilises as NH ₃ and NO _x in the manure management system
$EF_{4,j}$	= Emission factor for N ₂ O emissions from atmospheric deposition of on soils and water surfaces, [kg N- N ₂ O / (kg NH ₃ -N + NO _x -N volatilized)], estimated with site-specific, regional, or national data if such data is available. Otherwise, default values from 2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories (IPCC 2019), Volume 4, Chapter 11, Table 11.3 (Updated), page 11.26 (0.01 kg N ₂ O-

N/(kg NH₃-N +NO_x-N volatilised) (as per the IPCC 2019 Guideline, Chapter 11, Volume 4 EF₄ value was considered as 0.010)

EF_{5, j} = Emission factor for indirect emission of N₂O from runoff in kg N₂O-N/kg N, estimated with site-specific, regional, or national data if such data is available. Otherwise, default values from 2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories (IPCC 2019), Volume 4, Chapter 11, Table 11.3 (Updated), page 11.26 can be used (0.0075 kg N₂O-N/(kg N leaching/runoff) (as per the IPCC 2019 Guideline, Chapter 11, Volume 4 EF₅ value was considered as 0.0110)

Estimation of NEX_{LT,y}:

The procedure for the calculation of N₂O emissions from Manure Management is decided according to the Figure 10.4 of 2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories. Because there is no available country- specific Tier 3 methodology, Tier 2 livestock population characterization, and N₂O from manure management is not a key category, and there is no population data by high and low productivity system; the direct and indirect N₂O emissions are estimated using Tier 1 method and IPCC defaults. The Tier 1 approach states that “country-specific nitrogen excretion rates or excretion rates excretion rates developed by other countries that have livestock with similar characteristics can be used if available. In the case where such data country-specific data cannot be collected or derived, or appropriate data are not available from another country, the IPCC default nitrogen excretion rates presented in Table 10.19 can be used”. Because it is not possible to collect country-specific data or data from other countries that have similar livestock characteristics, the default IPCC values are used for this project. Because IPCC designates Turkey as a Middle East country the mean default values for Middle East are used. Necessary calculations are done using the Equation 10.30 in the applied IPCC Guideline.

Equation (9)

$$Nex_{(T,P)} = N_{rate(T,P)} * \frac{TAM_{(T,P)}}{1000} * 365$$

Where:

N_(T,P) = Annual N excretion for livestock category T, kg N animal-1 yr-1 (production level P if using a Tier 1 approach)

N_{rate(T,P)} = Default N excretion rate, kg N (1000 kg animal mass) -1 day-1 for animal category T (and production level P, if using a Tier 1 (see Table 10.19)

TAM_(T,P) = Typical animal mass for livestock category T, kg animal-1

P = Productivity class, high or low, to be considered if using the Tier 1 approach

Accordingly,

$$BE_{AW,N2O,y} = GWP_{N2O} * CF_{N2O-N,N} * 10^{-3} * (E_{N2O,D,y} + E_{N2O,ID,y})$$

$$BE_{AW,N2O,y} = 265 * 1.57 * 10^{-3} * (287 + 4,615)$$

$$BE_{AW,N2O,y} = 2,041 \text{ tCO}_2\text{e/year}$$

4.1.a.2 CO₂ emissions from electricity and heat within the project boundary $BE_{\frac{elec}{heat},y}$

In accordance with AM0073 “GHG emission reductions through multi-site manure collection and treatment in a central plant”, version 01.0, CO₂ emissions from electricity and/or heat used in the baseline is calculated as shown in equation (10) below.

Equation (10)

$$BE_{\frac{elec}{heat},y} = EG_{Bl,y} \times CEF_{Bl,elec,y} + EG_{d,y} \times CEF_{grid} + HG_{Bl,y} \times CEF_{Bl,therm,y}$$

Where:

$BE_{\frac{elec}{heat},y}$ = Baseline CO₂ emissions from electricity and/or heat used in the baseline, in tCO₂e/year

$EG_{Bl,y}$ = Amount of electricity in the year y that would be consumed in the absence of the project activity (MWh) for operating all AWMs facilities, accounted as zero (please see Section 3.3)

$CEF_{Bl,elec,y}$ = Carbon emissions factor for electricity consumed at the project site in the absence of the project activity (tCO₂e /MWh), accounted as zero (please see Section 3.3)

$EG_{d,y}$ = Amount of electricity generated utilizing the biogas collected during project activity and exported to the grid during the year y (MWh)

CEF_{grid} = Carbon emissions factor for the grid in the project scenario (tCO₂e /MWh)

$HG_{BL,y}$ = Quantity of thermal energy that would be consumed in year y in the absence of the project activity (MJ) using fossil fuel for operating all AWMSs, accounted as zero (please see Section 3.3)

$CEF_{BL,therm,y}$ = CO₂ emissions intensity for thermal energy generation (tCO_{2e} / MJ), accounted as zero (please see Section 3.3)

In accordance with the methodology of AM0073, the baseline emissions with electricity generation in year y ($CEF_{BL,therm,y}$) shall be calculated using the “Tool to calculate baseline, project and/or leakage emissions from electricity consumption” However, the Tool became inactive in 2015 and is recently replaced with the “Methodological tool – Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation”, version 03.0³¹.

Generic approach

As per “Methodological tool: Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation”, version 03.0, in the generic approach, baseline emissions from consumption of electricity are calculated based on the quantity of electricity consumed, an emission factor for electricity generation and a factor to account for transmission losses, as follows:

Equation (11)

$$BE_{EC,y} = \sum_k EC_{BL,k,y} \times EF_{EL,k,y} \times (1 + TDL_{k,y})$$

Where:

$BE_{EC,y}$ = Baseline emissions from electricity consumption in year y (tCO₂ / yr)

$EC_{BL,k,y}$ = Quantity of electricity that would be consumed by the baseline electricity consumption source k in year y (MWh/yr)

$EF_{EL,k,y}$ = Emission factor for electricity generation for source k in year y (tCO₂/MWh)

$TDL_{k,y}$ = Average technical transmission and distribution losses for providing electricity to source k in year y

k = Sources of electricity consumed by the grid

Determination of the emission factor for electricity generation ($EF_{EL,k,y}$)

³¹ https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-05-v1.pdf/history_view

The emission factor for the electricity generation $EF_{EL,k,y}$ will be determined in accordance with Scenario A (Electricity consumption from the grid) as defined under “Methodological tool: Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation”, version 03.0.

According to the Tool, the “combined margin” emission factor is calculated through the following six steps:

Step (1): Identify the relevant electricity system;

Step (2): Choose whether to include off-grid power plant in the project electricity system (optional);

Step (3): Select a method to determine the operating margin (OM);

Step (4): Calculate the operating margin emission factor according to the selected method;

Step (5): Calculate the build margin (BM) emission factor; Step (6): Calculate the combined margin (CM) emission factor.

Step 1: Identify the relevant electricity system

According to the applicable “Tool to calculate the emission factor for an electricity system”, version 07.0, the relevant project electricity system must be identified by the project participants. a project electricity system must be defined by the spatial extent of the power plants that are physically connected through transmission and distribution lines to the project activity, and that can be dispatched without significant transmission constraints.

Step 2: Choose whether to include off-grid power plants in the project electricity system (optional)

According to the applicable “Tool to calculate the emission factor for an electricity system”, version 07.0, project participants may choose between the following two options to calculate the operating margin and build margin emission factor:

Option I: Only grid power plants are included in the calculation.

Option II: Both grid power plants and off-grid power plants are included in the calculation.

The project participants choose Option I and therefore only grid power plants are included in the calculation.

Step 3: Select a method to determine the operating margin (OM)

According to the applicable “Tool to calculate the emission factor for an electricity system”, version 07.0, the operating margin emission factor ($EF_{grid,OM,y}$) is based on one of the following methods:

- a) Simple OM; or
- b) Simple adjusted OM; or
- c) Dispatch data analysis OM; or
- d) Average OM.

Step 4: Calculate the operating margin emission factor according to the selected method

The simple OM emission factor is calculated as the generation-weighted average CO₂ emissions per unit net electricity generation (tCO₂/MWh) of all generating power plants serving the system, not including low-cost/must-run power plants/units.

The simple OM may be calculated by one of the following two options:

Option A: Based on the net electricity generation and a CO₂ emission factor of each power unit; or

Option B: Based on the total net electricity generation of all power plants serving and the fuel types and total fuel consumption of the project electricity system.

Ministry of Energy and Natural resources of Turkey has published the data for OM using simple OM calculation³²

$$EF_{grid,OM,y} = 0.7424$$

For this approach (simple OM) to calculate the operating margin, the subscript m refers to the power plants/units delivering electricity to the grid, not including

Step 5: Calculate the build margin (BM) emission factor

In terms of vintage of data, project participants can choose between one of the following two options:

Option 1: for the first crediting period, calculate the build margin emission factor ex ante based on the most recent information available on units already built for sample group m at the time of PD submission to the DoE for validation.

Option 2: for the first crediting period, the build margin emission factor shall be updated annually, ex post, including those units built up to the year of registration of the project activity or, if

³²<https://enerji.gov.tr//Media/Dizin/EVCED/tr/%C3%87evreVe%C4%B0klim/%C4%B0klimDe%C4%9Fi%C5%9Fikli%C4%9Fi/TUESEmisyonFktr/Belgeler/Bform2020.pdf>

information up to the year of registration is not yet available, including those units built up to the latest year for which information is available.

For the proposed project activity, the project participants choose Option 1 in terms of availability of vintage of data.

Based on 2019 data, Ministry of Energy and Natural resources of Turkey has calculated build margin and published the data:

$$EF_{grid,BM,y} = 0.3680$$

Step 6: Calculate the combined margin emissions factor

According to the applicable “Tool to calculate the emission factor for an electricity system”, version 07.0, the calculation of the combined margin (CM) emission factor ($EF_{grid,CM,y}$) is based on one of the following methods:

The combined margin emissions factor is calculated as follows:

Equation (8)

$$EF_{grid,CM,y} = EF_{grid,OM,y} \times w_{OM} + EF_{grid,BM,y} \times w_{BM}$$

Where:

$EF_{grid,BM,y}$ = Build margin CO₂ emission factor in year y (tCO₂/MWh)

$EF_{grid,CM,y}$ = Combined margin CO₂ emission factor in year y (tCO₂/MWh)

$EF_{grid,OM,y}$ = Operating margin CO₂ emission factor in year y (tCO₂/MWh)

w_{OM} = Weighted of operating margin emissions factor (per cent)

w_{BM} = Weighted of build margin emissions factor (per cent)

As per the applicable “Tool to calculate the emission factor for an electricity system”, version 07.0, the default weights for the operating margin and build margin emission factors for the proposed project activity is defined as:

$$w_{OM} = 0.50$$

$$w_{BM} = 0.50$$

Based on 2019 data, Ministry of Energy and Natural resources of Turkey has calculated combined margin and published the data:

$$EF_{grid,CM,y} = 0.5552^{33}$$

Accordingly,

$$BE_{EC,y} = 90,000 \text{ MWh/year} \times 0.5552 \text{ tCO}_2\text{e/year} \times (1+0.096)$$

$$BE_{EC,y} = 54,764 \text{ tCO}_2\text{e/year}$$

4.2 Project Emissions

In accordance with AM0073 “GHG emission reductions through multi-site manure collection and treatment in a central plant”, version 01.0, the project activity might include one or a combination of technologies to treat manure. For example, the effluent mix might be first treated in an anaerobic digester/reactor and then treated waste might be further processed using an aerobic pond. Each processing step is referred to as a treatment stage. Project emissions are estimated as follows:

Equation (12)

$$PE_y = PE_{AD,y} + PE_{Aer,y} + PE_{Comp,y} + PE_{N_2O,y} + PE_{Pl,y} + PE_{flare,y} + PE_{elec_{heat},y} + PE_{CO_2,Trans,y} + PE_{storage,y}$$

Where:

PE_y = Project emissions (tCO₂e/yr)

$PE_{AD,y}$ = Leakage from treatment stage that captures methane (tCO₂e /yr)

$PE_{Aer,y}$ = Methane emissions from the aerobic treatment stage (tCO₂e /yr)

$PE_{Comp,y}$ = Total project emissions due to composting (tCO₂e /yr)

$PE_{N_2O,y}$ = Nitrous oxide emission from project treatment system (tCO₂e /yr)

$PE_{Pl,y}$ = Physical leakage of emissions from biogas network to flare the captured methane or supply to the facility where it is used for heat and/or electricity generation (tCO₂e /yr)

$PE_{flare,y}$ = Project emissions from flaring of the residual gas stream (tCO₂e /yr)

$PE_{elec_{heat},y}$ = Project emissions from use of heat and/or electricity in the project case (tCO₂e /yr)

³³Obtained from the most recent official national grid emission factor data published by the Ministry of the Energy and Natural Resources in Turkey
<https://enerji.gov.tr//Media/Dizin/EVCED/tr/%C3%87evreVe%C4%B0klim/%C4%B0klimDe%C4%9Fi%C5%9Fikli%C4%9Fi/TUES EmisyonFktr/Belgeler/Bform2020.pdf>

$PE_{CO_2,Trans,y}$ = Project emissions from manure road transportation (tCO₂e /yr)

$PE_{storage,y}$ = Project emissions from manure storage (tCO₂e /yr)

Since the proposed project activity does not consist of aerobic AWMS treatment, $PE_{Aer,y}$ shall be accounted as “0”. In addition, for the proposed project does not involve composting, $PE_{Comp,y}$ shall also be accounted as “0”. Moreover, since the manure is not stored in outdoor open storage tanks more than 24 hours, $PE_{storage,y}$ shall be accounted as “0” too. Accordingly, the equation (12) above could be simplified as shown in the equation (13) below.

Equation (13)

$$PE_y = PE_{AD,y} + PE_{N_2O,y} + PE_{Pl,y} + PE_{flare,y} + PE_{elec_{heat},y} + PE_{CO_2,Trans,y}$$

4.2.a Methane emissions from AWMS where gas is captured $PE_{AD,y}$

In accordance with AM0073 “GHG emission reductions through multi-site manure collection and treatment in a central plant”, version 01.0, IPCC guidelines specify physical leakage from anaerobic digesters as being 15% of total biogas production. Where project participants use lower values for percentage of physical leakage, they should provide measurements or other source of evidence proving that this lower value is appropriate for the project.

Ex ante leakage to be reported in the PD will be estimated using the equation 14 below, with a leakage factor of 0.15.

Equation (14)

$$PE_{AD,y} = GWP_{CH_4} \times \rho_{CH_4,n} \times \frac{LF_{AD}}{(1 - LF_{AD})} \times 10^{-3} \times \sum_{h=1}^{8760} (FV_{RG,h} \times fv_{CH_4,RG,h})$$

Where:

$PE_{AD,y}$ = Leakage from AWMS systems that capture’s methane in tCO₂e /yr

GWP_{CH_4} = Global Warming Potential (GWP) of CH₄

$\rho_{CH_4,n}$ = Density of methane at normal (at room temperature 20° C and 1 atm pressure) conditions (6.7x10⁻⁴ t/m³)

$FV_{RG,h}$ = Volumetric flow rate of the captured biogas in dry basis at normal conditions in hour h (m³/h)

$fv_{CH_4,RG,h}$ = Volumetric fraction of methane in the captured biogas on dry basis in hour h (fraction)

LF_{AD} = Methane leakage from anaerobic digesters/reactor, default of 0.15

Accordingly,

$$PE_{AD,y} = 28 \times 0.00067 \times \frac{0.15}{(1 - 0.15)} \times 10^{-3} \times (1,272.70 \times 60\% \times 8760)$$

$$PE_{AD,y} = 22.15 \text{ tCO}_2\text{e/yr}$$

4.2.b Physical Leakage from distribution network of the captured methane in PE_{PL}

In accordance with AM0073 “GHG emission reductions through multi-site manure collection and treatment in a central plant”, version 01.0, this refers to leaks in the biogas system from the biogas pipeline delivery system. The sum of the quantities of captured methane fed to the flare, to the power plant and to the boiler (measures as per the monitoring plan) must be compared annually with the total methane generated as measured by meter at the outlet of the methane generating digester. The difference between the monitored value of methane generated and that consumed in flare/electricity generation/heat shall be accounted as leakage from the pipelines.

In the case where biogas is just flared and the pipeline from collection point to flare is short (i.e., less than 1 km, and for on-site delivery only), one flow meter can be used. In such cases the physical leakage may be considered as zero.

4.2.c Project emissions from heat use and electricity use PE_{elec/heat}

In accordance with AM0073 “GHG emission reductions through multi-site manure collection and treatment in a central plant”, version 01.0, project emissions from heat use and electricity use (PE_{elec/heat}) are calculated as follows:

Equation (15)

$$PE_{\frac{elec}{heat},y} = PE_{Elec,y} + \sum_j PE_{heat,j,y}$$

Where:

$PE_{Elec,y}$ = Are the emissions from consumption of electricity in the project case. The project emissions from electricity consumption ($PE_{Elec,y} = PE_{EC,y}$) will be calculated following the latest version of “Tool to calculate baseline, project and/or leakage emissions from electricity consumption”

$PE_{heat,j,y}$ = Are the emissions from consumption of heat in the project case. The project emissions from fossil fuel combustion ($PE_{heat,j,y} = PE_{FC,j,y}$) will be calculated following the latest version of “Tool to calculate project or leakage CO₂ emissions from fossil fuel combustion”. For this purpose, the processes j in the tool

corresponds to all fossil fuel combustion in the plant established as part of the project activity, as well as any other on-site fuel combustion for the purposes of the project activity.

It is noted that “Tool to calculate baseline, project and/or leakage emissions from electricity consumption” became inactive in 2015 and is recently replaced with the “Methodological tool – Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation”, version 03.0. In accordance with the Tool, in the generic approach, project emissions from consumption of electricity are calculated based on the quantity of electricity consumed, an emission factor for electricity generation and a factor to account for transmission losses, as follows:

Equation (16)

$$PE_{EC,y} = \sum_j EC_{PJ,j,y} \times EF_{EF,j,y} \times (1 + TDL_{j,y})$$

Where:

$PE_{EC,y}$	= Project emissions from electricity consumption in year y (tCO ₂ / yr)
$EC_{PJ,j,y}$	= Quantity of electricity consumed by the project electricity consumption source j in year y (MWh/yr)
$EF_{EF,j,y}$	= Emission factor for electricity generation for source j in year y (tCO ₂ /MWh)
$TDL_{j,y}$	= Average technical transmission and distribution losses for providing electricity to source j in year y
j	= Sources of electricity consumption in the project

Accordingly,

$$PE_{EC,y} = 2,504 \frac{MWh}{yr} \times 0.5552 tCO_2/MWh \times (1 + 0.096)$$

$$PE_{EC,y} = 1,524 tCO_2e/year$$

4.2.d Project emissions from flaring

In the proposed project activity, an enclosed flare will be installed. The flare is installed for the case of emergency and malfunction in the plant. Thus, $PE_{flare,y}$ is considered as zero.

Nevertheless, the monitoring plan of the facility shall measure the quantity of captured methane fed to the flare. If such measurement is positive, the emissions from flaring the biogas shall be determined following the procedure described in the methodological tool project emissions from flaring, version 03.0.

Equation (17)

$$PE_{flare,y} = GWP_{CH_4} \times \sum_{m=1}^{525600} F_{CH_4,RG,m} \times (1 - \eta_{flare,m}) \times 10^{-3}$$

Where:

$PE_{flare,y}$ = Project emissions from flaring of the residual gas in year y (tCO₂e)

GWP_{CH_4} = Global warming potential of methane valid for the commitment period (tCO₂e /tCH₄)

$F_{CH_4,RG,m}$ = Mass flow of methane in the residual gas in the minute m (kg)

$\eta_{flare,m}$ = Flare efficiency in minute m

4.2.e Project emissions from road transportation

In accordance with AM0073 “GHG emission reductions through multi-site manure collection and treatment in a central plant”, version 01.0, the project emissions from manure transportation from the collection points to the central treatment plant are to be calculated using distance travelled by trucks and the fuel emission factor, as follows:

Equation (18)

$$PE_{CO_2,Trans,y} = \left\{ \sum_i (N_{vehicles,i,y} \times Dist_{i,y} \times FC_{i,f}) \times \left[\sum_f NCV_f \times EF_{CO_2,f} \right] \right\}$$

Where:

$PE_{CO_2,Trans,y}$ = Project emissions from manure road transportation in tCO₂e /yr

$N_{vehicles,i,y}$ = Number of trips of vehicle types i used for transportation, with similar loading capacity

$Dist_{i,y}$ = Average distance per trip travelled by transportation vehicles types i during the year y (km)

$FC_{i,f}$	= Specific consumption of fuel type f in volume or mass unites per km for vehicle type i
NCV_f	= Net calorific value of fuel type f in TJ per volume or mass units
$EF_{CO_2,f}$	= CO ₂ emission factor of the fossil fuel type f used in transportation vehicles, (tCO ₂ e /TJ)

Emissions arising from the road transportation of treated manure shall be calculated as described above. Such emission shall be considered as project emissions if the final destiny and itinerary between the treatment plant are included in the project boundary. Otherwise such emission shall be considered as leakage ($LE_{CO_2,Trans,y}$), which shall be calculated in the same manner as depicted above.

In the equation, $\sum_i (N_{vehicles,i,y} \times Dist_{i,y} \times FC_{i,f})$ is the part that the methodology is calculating the total consumption of fuel type f in volume per year from manure road transportation. Since the PP is actively recording the data on the total consumption of diesel fuel f in volume by the vehicles, $\sum_i (N_{vehicles,i,y} \times Dist_{i,y} \times FC_{i,f})$ shall be taken from the PP's accounting records. For ex-ante calculation, this value is taken from the feasibility report.

Accordingly,

$$PE_{CO_2,Trans,y} = 98,550 \times 43.3 \times 74.8 \div 10^6$$

$$PE_{CO_2,Trans,y} = 271.3 \text{ tCO}_2\text{e/year}$$

4.2.f N₂O emissions from the central treatment plant (PE_{N2O,y})

In accordance with AM0073 “GHG emission reductions through multi-site manure collection and treatment in a central plant”, version 01.0, Nitrous oxide emission from project treatment system (PE_{N2O,y}) is calculated as follows:

Equation (19)

$$PE_{N2O,y} = GWP_{N2O} * CF_{N2O-N,N} * 10^{-3} * (E_{N2O,D,y} + E_{N2O,ID,y})$$

Equation (20)

$$E_{N2O,D,y} = \sum_n EF_{N2O,D,n} * \sum_{m=1}^{12} (Q_{EM,m} * [N]_{EM,m})$$

$$E_{N_2O,D,y} = 0 * \sum_{m=1}^{12} (24,356.7 * 0.09)$$

$$E_{N_2O,D,y} = 0$$

Equation (21)

$$E_{N_2O,ID,y} = EF_{N_2O,ID} * \sum_n F_{gasm;j} * \sum_{m=1}^{12} (Q_{EM,m} * [N]_{EM,m})$$

$$E_{N_2O,ID,y} = 52.61 \text{ kg N}_2\text{O-N/year}$$

Where:

$PE_{N_2O,y}$	= Annual project N ₂ O emissions in tCO ₂ e/yr
GWP_{N_2O}	= Global Warming Potential (GWP) for N ₂ O
CF_{N_2O-N-N}	= Conversion factor N ₂ O-N to N ₂ O (44/28)
$E_{N_2O,D,y}$	= Direct N ₂ O emission in kg N ₂ O-N/year
$E_{N_2O,ID,y}$	= Indirect N ₂ O emission in kg N ₂ O-N/year
$EF_{N_2O,D,n}$	= Direct N ₂ O emission factor for the treatment stage n of the central treatment plant in kg N ₂ O-N/kg N (estimated with site-specific, regional, or national data if such data is available. Otherwise use default EF3 in volume 4, chapter 10, table 10.21 in IPCC 2006 guidelines)
$Q_{EM,m}$	= Monthly volume of the effluent mix entering the central plant (m ³ /month)
$[N]_{EM,m}$	=Monthly total nitrogen concentration in the effluent mix (kg N/m ³) entering the treatment plant
$EF_{N_2O,ID,n}$	=Indirect N ₂ O emission factor for N ₂ O emissions from atmospheric deposition of nitrogen on soils and water surfaces, kg N ₂ O-N/kg NH ₃ -N and NO _x -N emitted, estimated with site-specific, regional, or national data if such data is available. Otherwise, default values for EF ₄ from table 11.3, chapter 11, volume 4 of IPCC 2006 guidelines can be used.
$F_{gasm;j}$	=Percent of total nitrogen that volatilises as NH ₃ and NO _x in the treatment stage j

Accordingly,

$$PE_{N_2O,y} = GWP_{N_2O} * CF_{N_2O-N,N} * 10^{-3} * (E_{N_2O,D,y} + E_{N_2O,ID,y})$$

$$PE_{N_2O,y} = 265 * 1.57 * 10^{-3} * (0 + 52.61)$$

$$PE_{N_2O,y} = 21.91 \text{ t CO}_2\text{e/year}$$

4.3 Leakage

Leakage covers the emissions from land application of treated manure, outside the project boundary. The project activity includes collecting the manure, treating it with anaerobic digesters and spreading the effluent in fields after treatment. These emissions are estimated as net of those released under project activity and those released in the baseline scenario. In accordance with AM0073 Version 01.0, net leakage of N₂O and CH₄ are only considered if, they are positive. CO₂ emissions due to the road transportation of sludge or treated effluent outside the project boundary are also considered as leakage. Such emissions are calculated in the same as depicted in the project emissions section.

Since there is no land application of the effluent, the leakage is zero.

4.4 Estimated Net GHG Emission Reductions and Removals

Emission reductions are calculated as follows:

$$ER_y = BE_y - PE_y$$

Where:

ER_y = Emission reductions in year y (tCO₂e/yr)

BE_y = Baseline reductions in year y (tCO₂e/yr)

PE_y = Project reductions in year y (tCO₂e/yr)

$$ER_y = BE_y - PE_y$$

$$ER_y = 107,374 - 1,839 = 105,535 \text{ tCO}_2\text{e/year}$$

Year	Estimated baseline emissions or removals (tCO ₂ e)	Estimated project emissions or removals (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated net GHG emission reductions or removals (tCO ₂ e)
29/06/2021-31/12/2021	54,717	937	0	53,780

2022	107,374	1,839	0	105,535
2023	107,374	1,839		105,535
2024	107,374	1,839	0	105,535
2025	107,374	1,839	0	105,535
2026	107,374	1,839	0	105,535
2027	107,374	1,839	0	105,535
01/01/2028-28/06/2028	52,798	902	0	51,756
Total	751,619	12,872	0	738,535

5 MONITORING

5.1 Data and Parameters Available at Validation

Data / Parameter	D _{CH4}
Data unit	t/m ³
Description	Density of methane at normal (at room temperature 20°C and 1 atm pressure) conditions
Source of data	Approved Large-Scale Methodology of AM0073: GHG emission reductions through multi-site manure collection and treatment in a central plant, Version 1.0
Value applied	0.00067
Justification of choice of data or description of measurement methods and procedures applied	0.00067 t/m ³ at room temperature 20°C and 1 atm pressure
Purpose of Data	Used in; calculation of baseline emissions
Comments	-

Data / Parameter	GWP _{CH4}
Data unit	tCO ₂ e/tCH ₄
Description	Global Warming Potential for CH ₄

Source of data	IPCC Fifth Assessment Report, Synthesis Report, page 87
Value applied	28
Justification of choice of data or description of measurement methods and procedures applied	Default value from IPCC is used as per the applied methodology and as per Version 4.1 of the VCS Standard
Purpose of Data	Used in • Calculation of baseline emissions • Calculation of project emissions
Comments	In Version 4.1 of the VCS Standard, an update to VCS rules requires use of global warming potential values from the IPCC Fifth Assessment Report (updating from the Fourth Assessment Report) for calculating greenhouse gas (GHG) emission reductions. This update is consistent with the Paris Agreement and other carbon crediting programs.

Data / Parameter	$EF_{CO2,grid,y} = EF_{EF,j,y}$
Data unit	tCO ₂ e/MWh
Description	Emission factor for the Turkish National Grid
Source of data	Published data from The Ministry of the Energy and Natural Resources in Turkey
Value applied	0.5552
Justification of choice of data or description of measurement methods and procedures applied	Obtained from the most recent official national grid emission factor data published by the Ministry of the Energy and Natural Resources in Turkey
Purpose of Data	Used in • Calculation of baseline emissions • Calculation of project emissions
Comments	-

Data / Parameter	TDL _{j,y}
Data unit	-

Description	Average technical transmission and distribution losses for providing electricity to source j in year y
Source of data	Calculated from the Data Provided by Turkish Electricity Transmission Corporation (TEİAŞ) ³⁴ in accordance with Methodological Tool: Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation, Version 3.0
Value applied	0.096
Justification of choice of data or description of measurement methods and procedures applied	The annual average value based on the most recent data available within Turkey is used from the data provided by TEİAŞ.
Purpose of Data	Used in Calculation of project emissions
Comments	-

Data / Parameter	LF_{AD}
Data unit	Fraction
Description	Default Methane leakage from anaerobic digesters/reactor
Source of data	Approved Large-Scale Methodology of AM0073: GHG emission reductions through multi-site manure collection and treatment in a central plant, Version 1.0
Value applied	0.15
Justification of choice of data or description of measurement methods and procedures applied	
Purpose of Data	Used in Calculation of project emissions
Comments	-

Data / Parameter	$EF_{CO2,f}$
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³⁴ <https://webim.teias.gov.tr/file/5999ad34-fda7-4c1e-8496-e917716bad4e?download>

Data unit	Fraction (tCO ₂ e/TJ)
Description	CO ₂ emission factor of the fossil fuel type f used in transportation vehicles
Source of data	2006 IPCC Guidelines for National Greenhouse Gas Inventories, Volume 2, Chapter 1, Table 1.4, page 1.23 - Gas/Diesel Oil - Upper Limits – Conservative
Value applied	74.8
Justification of choice of data or description of measurement methods and procedures applied	
Purpose of Data	Used in Calculation of project emissions
Comments	-

Data / Parameter	NCV _f
Data unit	(Tj/Gg)
Description	Net calorific value of fuel type f in TJ per volume or mass units
Source of data	2006 IPCC Guidelines for National Greenhouse Gas Inventories, Volume 2, Chapter 1, Table 1.2, page 1.18 - Gas/Diesel Oil - Upper Limits – Conservative
Value applied	43.3
Justification of choice of data or description of measurement methods and procedures applied	
Purpose of Data	Used in Calculation of project emissions
Comments	-

Data / Parameter	VS _{default}
Data unit	kg/hd/day

Description	Annual volatile solid for livestock LT entering all AWMS [on a dry matter weight basis (kg-VS-dm/animal/day)]
Source of data	IPCC 2006, Table 10A-4 to 10A-9, Chapter 10, Volume 4
Value applied	Cow: 1.9 Cattle: 1.5 Chicken Layer: 0.02 Chicken Broiler: 0.01
Justification of choice of data or description of measurement methods and procedures applied	-
Purpose of Data	Used in Calculation of baseline emissions
Comments	-

Data / Parameter	W_{default}
Data unit	kg
Description	Average animal weight
Source of data	2019 refinement of 2006 IPCC vol 4 chapter 10 Table 10.A.5
Value applied	Cow: 510 Cattle: 491 Chicken Layer: 1.7 Chicken Broiler: 1
Justification of choice of data or description of measurement methods and procedures applied	-
Purpose of Data	Used in Calculation of baseline emissions
Comments	-

5.2 Data and Parameters Monitored

Data / Parameter	MCF _j
Data unit	Fraction
Description	Annual methane conversion factor (MCF) for the baseline animal manure management system j
Source of data	Obtained from 2019 Refinement to 2006 IPCC Guidelines for National Greenhouse Gas Inventories (IPCC 2019), Volume 4, Chapter 10, Table 10.17 (Updated), page 67 ³⁵
Description of measurement methods and procedures to be applied	For the project location, Malatya province in Turkey, Mean annual temperature is 13.7 ³⁶ °C ; Mean annual precipitation (MAP) is 385.8³⁷; Potential Evapotranspiration (PET) is 702³⁸ Hence, in accordance with Information Required to Determine Climate Zones According to Chapter 3 of Volume 5 Current Guideline of IPCC 2019, page 10.129, Malatya is considered as Warm Temperate Dry as its climate zone. j is Uncovered anaerobic lagoon. Thus, MCF is accounted accordingly.
Frequency of monitoring/recording	-
Value applied	Cow: 0.76 Cattle: 0.76 Egg chicken-Layer (wet): 0.73 Broiler: 0.015
Monitoring equipment	-
QA/QC procedures to be applied	

³⁵ https://www.ipcc-nggip.iges.or.jp/public/2019rf/pdf/4_Volume4/19R_V4_Ch10_Livestock.pdf

³⁶ <https://www.mgm.gov.tr/veridegerlendirme/il-ve-ilceler-istatistik.aspx?k=H&m=MALATYA>

³⁷ <https://www.mgm.gov.tr/veridegerlendirme/il-ve-ilceler-istatistik.aspx?k=H&m=MALATYA>

³⁸ http://tucaum.ankara.edu.tr/wp-content/uploads/sites/280/2015/08/tucaum4_4.pdf

Purpose of Data	Used in Calculation of baseline emissions
Calculation method	-
Comments	-

Data / Parameter	B _{0,LT}
Data unit	m ³ CH ₄ /kgVS
Description	Maximum methane producing potential of the volatile solid generated for animal type 'LT'
Source of data	IPCC 2019, Volume 4, Chapter 10, Table 10.16, page 10.66
Frequency of monitoring/recording	Annually
Value applied	For Other Region – High Productivity Systems (PS), Dairy Cattle: 0.24; Chicken Layer (Hens): 0.39 Chicken Broiler:0.36
Justification of choice of data or description of measurement methods and procedures applied	In accordance with the 2019 Refinement, Turkey is regarded under Middle East region.
Purpose of Data	Used in Calculation of baseline emissions
Comments	-

Data / Parameter	N _p
Data unit	Numbers
Description	Number of animals produced annually of type LT for the year y
Source of data	Project developer records
Description of measurement methods and	N _p used to calculate N _{LT,y} will be measured based on the annual baseline surveys that will be conducted by the project developer with the animal farm owners, whilst the farm owners' official records will be cross-checked. Here, it is important to note that all

procedures to be applied	the farms from which the biogas plant receives animal manure are managed under confined conditions, in accordance with the relevant Turkish laws and regulations. More importantly, in accordance with the relevant law and regulations, the farm owners are mandated to keep the track of each animal that they have at their farm. Each animal has their ear tags stating their names, ages, weights, vaccinations etc. These official records that need to be prepared and be submitted to the governmental offices.
Frequency of monitoring/recording	Annually
Value applied	Cow: 7,060 Cattle: 8,030 Egg chicken-Layer (wet): 260,000 Broiler: 29,205
Monitoring equipment	Annual baseline surveys conducted with the farm owners, whilst farmers' record being crossed-checked
QA/QC procedures to be applied	
Purpose of data	To calculate Baseline emission
Calculation method	
Comments	

Data / Parameter	Nda
Data unit	Numbers
Description	Number of days animal is alive in the farm in the year y
Source of data	Project developer records
Description of measurement methods and procedures to be applied	Nda used to calculate NLT,Y will be measured based on the annual baseline surveys that will be conducted by the project developer with the animal farm owners, whilst the farm owners' official records will be cross-checked. Here, it is important to note that all of the farms from which the biogas plant receives animal manure

	are managed under confined conditions, in accordance with the relevant Turkish laws and regulations. More importantly, in accordance with the relevant law and regulations, the farm owners are mandated to keep the track of each animal that they have at their farm. Each animal has their ear tags stating their names, ages, weights, vaccinations etc. These official records that need to be prepared and be submitted to the governmental offices.
Frequency of monitoring/recording	Annually
Value applied	365
Monitoring equipment	Annual baseline surveys conducted with the farm owners, whilst farmers' record being crossed-checked
QA/QC procedures to be applied	
Purpose of data	To calculate Baseline emission
Calculation method	
Comments	

Data / Parameter	$Q_{EM,m}$
Data unit	m ³ /month
Description	Monthly volume of the effluent mix entering the central treatment plant
Source of data	Feasibility report, project proponent records
Description of measurement methods and procedures to be applied	-
Frequency of monitoring/recording	Continuous monitoring
Value applied	24,356.7

Monitoring equipment	-
QA/QC procedures to be applied	-
Purpose of data	To calculate project emission
Calculation method	-
Comments	-

Data / Parameter	$\sum_i (N_{vehicles,i,y} \times Dist_{i,y} \times FC_{i,f})$
Data unit	Liters/year
Description	Average consumptions of fuel type f in volume per year from manure road transportation
Source of data	On-site measurement
Description of measurement methods and procedures to be applied	Log-book records
Frequency of monitoring/recording	Annually, based on daily records, monthly aggregation
Value applied	98,550
Monitoring equipment	Diesel Oil Bills Received from Oil Station(s)
QA/QC procedures to be applied	-
Purpose of data	Calculation of project emissions
Calculation method	

Comments	
Data / Parameter	$D_{ist,i,y}$
Data unit	km
Description	Average distance per trip travelled by transportation vehicles type i
Source of data	Project proponent (Feasibility report)
Description of measurement methods and procedures to be applied	-
Frequency of monitoring/recording	-
Value applied	-
Monitoring equipment	-
QA/QC procedures to be applied	-
Purpose of data	To calculate projects emissions
Calculation method	-
Comments	($N_{vehicles,i,y} \times Dist_{i,i,y} \times FC_{i,f}$) formulation (as mentioned above) refers to the total consumption of fuel type f in volume per year from manure road transportation and it is used in the CO ₂ emission from the transportation. Hence, this parameter is considered 1.

Data / Parameter	$N_{ist,i,y}$
Data unit	Number
Description	Number of trips of vehicles type i used for transportation, with similar loading capacity
Source of data	-

Description of measurement methods and procedures to be applied	-
Frequency of monitoring/recording	-
Value applied	1
Monitoring equipment	-
QA/QC procedures to be applied	-
Purpose of data	To calculate projects emissions
Calculation method	-
Comments	($N_{\text{vehicles},i,y} \times \text{Dist } i,i,y \times FC_{i,f}$) formulation (as mentioned above) refers to the total consumption of fuel type f in volume per year from manure road transportation and it is used in the CO ₂ emission from the transportation. Hence, this parameter is considered 1.

Data / Parameter	$FC_{i,f}$
Data unit	liter/km
Description	Specific consumption of fuel type f in volume or mass units per km for vehicle type i
Source of data	-
Description of measurement methods and procedures to be applied	-
Frequency of monitoring/recording	Daily
Value applied	1
Monitoring equipment	-

QA/QC procedures to be applied	-
Purpose of data	To calculate projects emissions
Calculation method	-
Comments	($N_{vehicles,i,y} \times Dist_{i,i,y} \times FC_{i,f}$) formulation (as mentioned above) refers to the total consumption of fuel type f in volume per year from manure road transportation and it is used in the CO ₂ emission from the transportation. Hence, this parameter is considered 1.

Data / Parameter	EG _{d,y}
Data unit	MWh/yr
Description	Quantity of net electricity generation supplied by the project plant/unit to the grid in year y
Source of data	Electricity meter / Bill generated
Description of measurement methods and procedures to be applied	This parameter will be monitored energy meter Class 0.5S
Frequency of monitoring/recording	Continuous measurements and monthly recorded.
Value applied	90,000
Monitoring equipment	This parameter will be monitored using energy meter Class 0.5S
QA/QC procedures to be applied	The monitored equipment will be calibrated periodically according with manufacturer's recommendations or local/international standard to ensure accuracy and data will be recorded and backed-up.

Purpose of data	Calculation of project emissions
Calculation method	
Comments	

Data / Parameter	ndy
Data unit	<i>Numbers</i>
Description	Number of days the central treatment plant was operational in year y
Source of data	Project developer records
Description of measurement methods and procedures to be applied	
Frequency of monitoring/recording	Annually
Value applied	365
Monitoring equipment	Shall be measured according to the weight scale in the entrance of the facility, along with the managerial records and the bills regarding electricity generation
QA/QC procedures to be applied	
Purpose of data	
Calculation method	To calculate Baseline emission
Comments	

Data / Parameter	W_{site}
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Data unit	kg
Description	Average animal weight of a defined population at the project site in kg
Source of data	Project proponent
Description of measurement methods and procedures to be applied	-
Frequency of monitoring/recording	Annual record from farm owners
Value applied	Cow: 591 Cattle: 491 Egg chicken-Layer (wet): 1.8 Broiler: 1.5
Monitoring equipment	-
QA/QC procedures to be applied	-
Purpose of data	Calculation of baseline emissions
Calculation method	-
Comments	-

Data / Parameter	$EC_{PJ,j,y}$
Data unit	MWh/yr
Description	Quantity of electricity consumed by the project electricity consumption source j in year y
Source of data	Electricity meter / Bill generated

Description of measurement methods and procedures to be applied	Bill generated
Frequency of monitoring/recording	Continuous measurements and monthly recorded.
Value applied	2,504
Monitoring equipment	Two-way electricity meters
QA/QC procedures to be applied	periodic calibration will be done of the meters as per agreement
Purpose of data	Calculation of project emissions
Calculation method	-
Comments	

Data / Parameter	$FV_{RG,h}$
Data unit	m ³ /hour
Description	Volumetric flow rate of the captured biogas in dry basis at normal conditions in hour h (m ³ /h)
Source of data	SCADA (Feasibility report for ex-ante)
Description of measurement methods and procedures to be applied	
Frequency of monitoring/recording	Continuously
Value applied	1,273
Monitoring equipment	Flowmeter
QA/QC procedures to be applied	-
Purpose of data	Calculation of project emissions
Calculation method	-
Comments	

Data / Parameter	f _{VCH4,RG,h}
Data unit	fraction
Description	Volumetric fraction of methane in the captured biogas on dry basis in hour h (fraction)
Source of data	SCADA (Feasibility report for ex-ante)
Description of measurement methods and procedures to be applied	
Frequency of monitoring/recording	Continuously
Value applied	60%
Monitoring equipment	Flowmeter
QA/QC procedures to be applied	-
Purpose of data	Calculation of project emissions
Calculation method	-
Comments	

5.3 Monitoring Plan

Description of the Monitoring Plan: In line with sample collection procedures described in the Annex 2 of the applied Meth., manure samples from each farm are separately collected by (bio)chemical engineers working at the laboratory on a daily basis. Samples are used for the necessary tests such as VS measurement. Samples are analyzed as soon as collected by the laboratory personnel who are qualified to perform/conduct such tests. If necessary, samples are stored at 4°C at the lab. The analyses results are, first, recorded by the personnel to lab logbooks in a hard copy format and then, to electronic records. The data recorded electronically is going to be archived during project plus five years as per the applied Meth. TOHMA Biomass Power Plant Project's controlled via the Automatic Control System (SCADA). This control system provides continuous monitoring, recording, and processing of the operating parameters (Temperature, Pressure etc.).

All devices and installations are calibrated as necessary. Furthermore, the SCADA operator monitors all equipment constantly. The control system displays the flowrate data as well as most of the important process variables, such as volumes of gas holders, and plant sections.

Data recorded from weighbridge, gas engines and pumps to digesters are used as source of data for Number of days the central treatment plant was operational in a year (ndy). Such data sources are indeed crucial for a process. If any of this equipment were to fail, it would lead to an operation shut down. Hence, monitoring and keeping record of such data is crucial for the biogas plant operation. The data recorded electronically is going to be archived during project plus five years as per the applied Meth.

In addition, for SDG contribution reporting, the project owners' records on weighbridge, employment and YEKDEM scheme records will be archived electronically during project plus five years.

Executive Summary

Methods for measuring, recording, storing, aggregating, collating, and reporting data and parameters:

Monitoring will be carried out by recording the necessary data on a spreadsheet and by using supporting documentation. All data used for the monitoring plan will be archived electronically and backed up every 24 hours online. These data will be kept for the full crediting period and a further five years after the end of the crediting period or the last issuance of carbon credits, whichever occurs later.

The amount of biogas combusted in the gas engines is monitored by conservatively monitoring the methane amount supplied to each of the engine units. The actual emission reductions from methane utilization in the combustion units will be calculated in a way that the risk of overestimation is minimized.

The biogas plant's computerized management and control system includes: the management of the automatic loading of the fermenters, including the management of the quantities; monitoring of level management in gas meters through the level indicators; monitoring and control of the heating system; alarm management of minimum and maximum levels and security; and monitoring of the data analysis of the biogas analysis. The electricity generation, given that it dispatches electricity to the National Grid, will be measured, recorded, stored, aggregated, collated, and reported following the procedure described by the authorities.

The operators of the methane recovery and electricity generation system are responsible for collecting all data monitored on-site. The operating and maintenance personnel are skilled technicians, with extensive experience in equipment operation, maintenance and calibration, and emergency procedures. Overall responsibility for the monitoring and maintenance of all required tasks and their adequate management lies with the project manager. Detailed roles and responsibilities of the relevant staff involved in VCS monitoring are set. Furthermore, this staff will receive relevant training, if required, to ensure that monitoring duties will be performed by trained staff.

Operational and Management Structure:

- Site manager – to control all activities (managing the budget and personnel, process controlling) in the plant, including the control unit of the plant, monitoring equipment and the staff. The site manager records the amount of diesel oil used by vehicles used to transport manure ($\sum (N_{\text{vehicles},i,y} \times \text{Dist } i,y \times \text{FC}_{i,f})$), daily. There is a local oil station from where all vehicles get diesel oil. The station raises daily invoices to the project based on daily consumption of oil by vehicles. Such data are first recorded by the site manager to the project's operation-based logbook, then, passed to electronic archive. The data recorded electronically is going to be archived during project plus five years as per the applied Meth. The site manager, together with person in charge of manure supply processes, is responsible to conduct baseline surveys with manure suppliers. Such surveys include not only number of animals produced annually (N_p) but also number of days animal is alive in the farm (N_{da}). These data provided by farm owners to the project proponents are crossed checked with official records kept by farm owners and submitted to the government. Surveys are kept in hard copy formats at the project site, then, passed to electronic records. The data recorded electronically is going to be archived during project plus five years as per the applied Meth.
- Site engineer – to control all the process and biological treatment in the plant – control of process parameters, following up lab tests, comment on lab analysis, biogas measurements and manure management.
- The electrical and mechanical operators– the continuous operation of the gas engines, for electricity generation – The electrical and mechanical operator at site is responsible to record the data from flow meter measuring flare efficiency ($\eta_{\text{flare},m}$) when flare is used. The operator records the data to project's electrical and mechanical-based logbook first, then archive it electronically. The data recorded electronically is going to be archived during project plus five years as per the applied Meth. In addition, these people are in charge of the biogas scrubbing unit which cleans the biogas before the electricity production. Both the site manager and the electrical and mechanical operators are in charge of recording the data on amount of electricity used from the national grid ($E_{CP,J,y}$). Two-way electricity meters are used to cross check the monthly bills raised by the officially designated company in charge of electricity distribution on behalf of the government. The bills, together with meters' data, are recorded both in hard and electronic copy formats. The data recorded electronically is going to be archived during project plus five years as per the applied Meth.
- Both the site manager and the electrical and mechanical operator are in charge of recording the data on amount of electricity generated and supplied to the national grid ($E_{Gd,y}$). The main and backup electricity meters are recording every KWe fed to the grid continuously. In accordance with the Turkish legislation titled 'The Measurement and Measuring Instrument Regulations, the meters shall be calibrated and tested periodically by the manufacturer every ten years. Data obtained from meters are recorded monthly to the project's operation-based logbook. More importantly, the official designated company raises official bills on the amount of electricity fed to the grid every month. Such bills are also used and crossed check with meters' data for $E_{Gd,y}$ value. Data is recorded in a hard copy format, and also electronically. The data recorded electronically is going

to be archived during project plus five years as per the applied Meth. The Scada operator– monitors all equipment and process variables on Scada System.

Quality Assurance and Quality Control:

- Quality control and quality assurance procedures will guarantee the quality of monitored data.
- All data will be archived electronically and backed up in every 24 hours online.
- All data collected as part of monitoring will be archived electronically and be kept at least for five years after the end of the last crediting period. Regarding the monitoring equipment of the methane recovery unit, maintenance and calibration will be performed in line with manufacturers' recommendations. With regards to the electricity meter, as mentioned above, it is a high accuracy measurement device/s and meets all relevant metrological requirements prescribed by the state authority. Procedures for maintenance of the meter will be conducted in accordance with national procedures and standards.

Storing Process Variables and Recording Operational Data:

All-important process variables and measurement values are recorded and stored o by the Scada operator, site engineer, and site manager. The process variables can be displayed as needed. Such variables are:

- Amount of manure entering the plant,
- Amount of biogas produced at the reactors,
- Amount of electricity consumed by the operation of the plant and amount of electricity generated.
- All other parameters that are needed to be verified by the DOE.

Emergency Preparedness:

Operation Manager ensures that the project personnel acquire the skills to carry out emergency tasks properly through capacity building practices and trainings. The whole staff receives health and safety trainings to minimize exposure to workplace hazards. At every shift, at least one trained technical personnel are present. All relevant information such as data files, maintenance, defect records and other monitoring related records are kept at a designated location in a transparent and organized manner.

6 ACHIEVED GHG EMISSION REDUCTIONS AND REMOVALS

6.1 Data and Parameters Monitored

Data / Parameter	N_{da}
Data unit	Numbers
Description	Number of days animal is alive in the farm in the year y
Value applied:	365
Comments	-

Data / Parameter	$N_{LT,y}$
Data unit	Numbers
Description	Annual average number of animals of type LT for the year y, expressed in numbers
Value applied:	Cow: 7,000 Cattle: 8,000 Egg chicken-Layer (wet): 245,000 Broiler: 25,000
Comments	-

Data / Parameter	$\Sigma(N_{vehicles,i,y} * Dist_{y,f} * FC_{i,f})$
Data unit	Liters/year
Description	Average consumptions of fuel type f in volume per year from manure road transportation

Value applied:	98,000
Comments	According to the statistical data of the project owner

Data / Parameter	Dist _{i,y}
Data unit	km
Description	Average distance per trip travelled by transportation vehicles type i
Value applied:	1
Comments	(N _{vehicles,i,y} × Dist _{i,y} × FC _{i,f}) formulation (as mentioned above) refers to the total consumption of fuel type f in volume per year from manure road transportation and it is used in the CO ₂ emission from the transportation. Hence, this parameter is considered 1.

Data / Parameter	N _{ist,i,y}
Data unit	Number
Description	Number of trips of vehicles type i used for transportation, with similar loading capacity
Value applied:	1
Comments	N _{vehicles,i,y} × Dist _{i,y} × FC _{i,f} formulation (as mentioned above) refers to the total consumption of fuel type f in volume per year from manure road transportation and it is used in the CO ₂ emission from the transportation. Hence, this parameter is considered 1.

Data / Parameter	FC _{i,f}
Data unit	liter/km
Description	Specific consumption of fuel type f in volume or mass units per km for vehicle type i
Value applied:	1
Comments	N _{vehicles,i,y} × Dist _{i,y} × FC _{i,f} formulation (as mentioned above) refers to the total consumption of fuel type f in volume per year from manure road transportation and it is used in the CO ₂ emission from the transportation. Hence, this parameter is considered 1.

Data / Parameter	W _{site}
Data unit	kg
Description	Average animal weight of a defined population at the project site in kg
Value applied:	Cow: 600 Cattle: 500 Egg chicken-Layer (wet): 1.9 Broiler: 1.4
Comments	-

Data / Parameter	EG _{d,y}
Data unit	MWh/year
Description	Quantity of net electricity generation supplied by the project plant/unit to the grid in year y
Value applied:	79,550

Comments	Quantity of net electricity (difference between the total quantity of electricity generated by the power plant/unit and the auxiliary electricity consumption) generation that is produced and fed into the grid as a result of the implementation of the project activity in year y
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Data / Parameter	ndy
Data unit	Numbers
Description	Number of days the central treatment plant was operational in year y
Value applied:	365
Comments	According to the operation log of the project owner

Data / Parameter	EC _{PJ,y}
Data unit	MWh/year
Description	Quantity of electricity consumed by the project electricity consumption in year y
Value applied:	85
Comments	According to the data shown in the electricity meter

Data / Parameter	FV _{RG,h}
Data unit	m ³ /hour
Description	Volumetric flow rate of the captured biogas in dry basis at normal conditions in hour h
Value applied:	2,000

Comments	-
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Data / Parameter	f _{VCH4,RG,h}
Data unit	Fraction
Description	Volumetric fraction of methane in the captured biogas on dry basis in hour h
Value applied:	90%
Comments	-

Data / Parameter	N
Data unit	Number
Description	Total number of farms
Value applied:	79
Comments	The contract have been signed with the farms so that continuous waste flow can be supplied

Data / Parameter	T
Data unit	°C
Description	Monthly average ambient temperature at the livestock farms included in the project boundary
Value applied:	13.7 ³⁹
Comments	Used to select the annual MCFj from IPCC 2006 Guideline

³⁹ <https://www.mgm.gov.tr/veridegerlendirme/il-ve-ilceler-istatistik.aspx?m=MALATYA>

Data / Parameter	MCF _i
Data unit	%
Description	Annual methane conversion factor (MCF) for the baseline animal manure management system j
Value applied:	Cow: 0.76 Cattle: 0.76 Egg chicken-Layer (wet): 0.73 Broiler: 0.015
Comments	

6.2 Baseline Emissions

The baseline emissions are:

- Emissions from cattle and chicken manure left to decay in anaerobic conditions at the farms;
- Emissions resulting from electricity generation.

The calculation of the baseline emissions for the proposed project activities are based on the methodology of AM0073 “Approved baseline and monitoring methodology – GHG emission reductions through multi-site manure collection and treatment in a central plant”, version 01.0.

In accordance with AM0073 “GHG emission reductions through multi-site manure collection and treatment in a central plant”, version 01.0, baseline emissions are calculated as the sum of CH₄ and N₂O emissions that would occur in the baseline animal waste treatment system and CO₂ emissions arising from heat and consumption.

According to the methodology AM0073, baseline emissions are calculated by using equation (1) and include the following sources:

$$BE_y = BE_{AW,y} + BE_{\frac{elec}{heat}y}$$

6.2.a.1. Emissions from animal waste treatment

As per AM0073 “GHG emission reductions through multi-site manure collection and treatment a central plant”, version 01.0, the baseline is the AWMS identified through the baseline selection procedure.

$$BE_{AW,y} = BE_{AW,CH_4,y} + BE_{AW,N_2O,y}$$

Methane emissions from animal waste treatment

In accordance with AM0073 “GHG emission reductions through multi-site manure collection and treatment in a central plant”, version 01.0, manure management system in the baseline could be based on different treatment systems and on one or more stages. Therefore:

Equation (3)

$$BE_{AW,CH_4,y} = GWP_{CH_4} \times \rho_{CH_4} \times \sum_{j,LT} (MCF_j \times B_{0,LT} \times N_{LT} \times VS_{LT,y} \times MS\%_{Bl,j})$$

N₂O emissions from manure management

$$BE_{AW,N_2O,y} = GWP_{N_2O} * CF_{N_2O-N,N} * 10^{-3} * (E_{N_2O,D,y} + E_{N_2O,ID,y})$$

$$E_{N_2O,D,y} = \sum_{j,LT} (EF_{N_2O,D,j} * NEX_{LT,y} * N_{LT,y} * MS\%_{Bl,j})$$

$$E_{N_2O,ID,y} = \sum_{j,LT} ((EF_{4,j} + EF_{5,j}) * NEX_{LT,y} * F_{gasm} * MS\%_{Bl,j})$$

6.2.a.2 CO₂ emissions from electricity and heat within the project boundary $BE_{elec,heat,y}$

In accordance with AM0073 “GHG emission reductions through multi-site manure collection and treatment in a central plant”, version 01.0, CO₂ emissions from electricity and/or heat used in the baseline is calculated as shown in equation below.

$$BE_{elec,heat,y} = EG_{Bl,y} \times CEF_{Bl,elec,y} + EG_{d,y} \times CEF_{grid} + HG_{Bl,y} \times CEF_{Bl,therm,y}$$

6.3 Project Emissions

In accordance with AM0073 “GHG emission reductions through multi-site manure collection and treatment in a central plant”, version 01.0, the project activity might include one or a combination of technologies to treat manure. For example, the effluent mix might be first treated in an anaerobic digester/reactor and then treated waste might be further processed using an aerobic pond. Each processing step is referred to as a treatment stage. Project emissions are estimated as follows:

Equation (12)

$$PE_y = PE_{AD,y} + PE_{Aer,y} + PE_{Comp,y} + PE_{N_2O,y} + PE_{Pl,y} + PE_{flare,y} + PE_{elec_{heat},y} + PE_{CO_2,Trans,y} + PE_{storage,y}$$

Where:

PE_y	= Project emissions (tCO ₂ e/yr)
$PE_{AD,y}$	= Leakage from treatment stage that captures methane (tCO ₂ e /yr)
$PE_{Aer,y}$	= Methane emissions from the aerobic treatment stage (tCO ₂ e /yr)
$PE_{Comp,y}$	= Total project emissions due to composting (tCO ₂ e /yr)
$PE_{N_2O,y}$	= Nitrous oxide emission from project treatment system (tCO ₂ e /yr)
$PE_{Pl,y}$	= Physical leakage of emissions from biogas network to flare the captured methane or supply to the facility where it is used for heat and/or electricity generation (tCO ₂ e /yr)
$PE_{flare,y}$	= Project emissions from flaring of the residual gas stream (tCO ₂ e /yr)
$PE_{elec_{heat},y}$	= Project emissions from use of heat and/or electricity in the project case (tCO ₂ e /yr)
$PE_{CO_2,Trans,y}$	= Project emissions from manure road transportation (tCO ₂ e /yr)
$PE_{storage,y}$	= Project emissions from manure storage (tCO ₂ e /yr)

Since the proposed project activity does not consist of aerobic AWMS treatment, $PE_{Aer,y}$ shall be accounted as “0”. In addition, for the proposed project does not involve composting, $PE_{Comp,y}$ shall also be accounted as “0”. Moreover, since the manure is not stored in outdoor

open storage tanks more than 24 hours, $PE_{storage,y}$ shall be accounted as “0” too. Accordingly, the equation (12) above could be simplified as shown in the equation (13) below.

Equation (13)

$$PE_y = PE_{AD,y} + PE_{N_2O,y} + PE_{Pl,y} + PE_{flare,y} + PE_{elec_{heat},y} + PE_{CO_2,Trans,y}$$

6.3.a Methane emissions from AWMS where gas is captured $PE_{AD,y}$

In accordance with AM0073 “GHG emission reductions through multi-site manure collection and treatment in a central plant”, version 01.0, IPCC guidelines specify physical leakage from anaerobic digesters as being 15% of total biogas production. Where project participants use lower values for percentage of physical leakage, they should provide measurements or other source of evidence proving that this lower value is appropriate for the project.

Ex ante leakage to be reported in the PD will be estimated using the equation 14 below, with a leakage factor of 0.15.

Equation (14)

$$PE_{AD,y} = GWP_{CH_4} \times \rho_{CH_4,n} \times \frac{LF_{AD}}{(1 - LF_{AD})} \times 10^{-3} \times \sum_{h=1}^{8760} (FV_{RG,h} \times fv_{CH_4,RG,h})$$

6.3.b Physical Leakage from distribution network of the captured methane in PE_{Pl}

In accordance with AM0073 “GHG emission reductions through multi-site manure collection and treatment in a central plant”, version 01.0, this refers to leaks in the biogas system from the biogas pipeline delivery system. The sum of the quantities of captured methane fed to the flare, to the power plant and to the boiler (measures as per the monitoring plan) must be compared annually with the total methane generated as measured by meter at the outlet of the methane generating digester. The difference between the monitored value of methane generated and that consumed in flare/electricity generation/heat shall be accounted as leakage from the pipelines.

In the case where biogas is just flared and the pipeline from collection point to flare is short (i.e., less than 1 km, and for on-site delivery only), one flow meter can be used. In such cases the physical leakage may be considered as zero.

6.3.c Project emissions from heat use and electricity use $PE_{elec/heat}$

In accordance with AM0073 “GHG emission reductions through multi-site manure collection and treatment in a central plant”, version 01.0, project emissions from heat use and electricity use ($PE_{elec/heat}$) are calculated as follows:

Equation (15)

$$PE_{\frac{elec}{heat},y} = PE_{Elec,y} + \sum_j PE_{heat,j,y}$$

Equation (16)

$$PE_{EC,y} = \sum_j EC_{PJ,j,y} \times EF_{EF,j,y} \times (1 + TDL_{j,y})$$

6.3.d Project emissions from flaring

In the proposed project activity, an enclosed flare will be installed. The flare is installed for the case of emergency and malfunction in the plant. Thus, $PE_{flare,y}$ is considered as zero. Nevertheless, the monitoring plan of the facility shall measure the quantity of captured methane fed to the flare. If such measurement is positive, the emissions from flaring the biogas shall be determined following the procedure described in the methodological tool project emissions from flaring, version 03.0.

Equation (17)

$$PE_{flare,y} = GWP_{CH_4} \times \sum_{m=1}^{525600} F_{CH_4,RG,m} \times (1 - \eta_{flare,m}) \times 10^{-3}$$

Where:

$PE_{flare,y}$ = Project emissions from flaring of the residual gas in year y (tCO₂e)

GWP_{CH_4} = Global warming potential of methane valid for the commitment period (tCO₂e / tCH₄)

$F_{CH_4,RG,m}$ = Mass flow of methane in the residual gas in the minute m (kg)

$\eta_{flare,m}$ = Flare efficiency in minute m

6.3.e Project emissions from road transportation

In accordance with AM0073 “GHG emission reductions through multi-site manure collection and treatment in a central plant”, version 01.0, the project emissions from manure transportation from the collection points to the central treatment plant are to be calculated using distance travelled by trucks and the fuel emission factor, as follows:

Equation (18)

$$PE_{CO_2,Trans,y} = \left\{ \sum_i (N_{vehicles,i,y} \times Dist_{i,y} \times FC_{i,f}) \times \left[\sum_f NCV_f \times EF_{CO_2,f} \right] \right\}$$

6.3.f N₂O emissions from the central treatment plant (PE_{N2O,y})

In accordance with AM0073 “GHG emission reductions through multi-site manure collection and treatment in a central plant”, version 01.0, Nitrous oxide emission from project treatment system (PE_{N2O,y}) is calculated as follows:

Equation (19)

$$PE_{N2O,y} = GWP_{N2O} * CF_{N2O-N,N} * 10^{-3} * (E_{N2O,D,y} + E_{N2O,ID,y})$$

Equation (20)

$$E_{N2O,D,y} = \sum_n EF_{N2O,D,n} * \sum_{m=1}^{12} (Q_{EM,m} * [N]_{EM,m})$$

Equation (21)

$$E_{N2O,ID,y} = EF_{N2O,ID} * \sum_n F_{gasm;j} * \sum_{m=1}^{12} (Q_{EM,m} * [N]_{EM,m})$$

6.4 Leakage

Leakage covers the emissions from land application of treated manure, outside the project boundary. The project activity includes collecting the manure, treating it with anaerobic digesters and spreading the effluent in fields after treatment. These emissions are estimated as net of those released under project activity and those released in the baseline scenario. In accordance with AM0073 Version 01.0, net leakage of N₂O and CH₄ are only considered if, they are positive.

CO₂ emissions due to the road transportation of sludge or treated effluent outside the project boundary are also considered as leakage. Such emissions are calculated in the same as depicted in the project emissions section.

Since there is no land application of the effluent, the leakage is zero.

6.5 Net GHG Emission Reductions and Removals

Verification period: From 29/06/2021 to 30/09/2022 (both days inclusive)

The emission reduction achieved by the project activity can be estimated ex ante as follows.

$$ER_{y,estimated} = BE_y - PE_y - LE_y$$

Where:

$ER_{y,estimated}$ = Estimated emission reduction from both methane destruction and grid displacement (tCO₂)

BE_y = Baseline emissions from both methane destruction and grid displacement (tCO₂)

PE_y = Project emissions from electricity or fossil fuel use (tCO₂)

LE_y = Leakage emissions from both methane destruction and grid displacement (tCO₂)

Year	Baseline emissions or removals (tCO ₂ e)	Project emissions or removals (tCO ₂ e)	Leakage emissions (tCO ₂ e)	Net GHG emission reductions or removals (tCO ₂ e)
29/06/2021-31/12/2021	51,615	203	0	51,412
01/01/2022-30/09/2022	75,481	297	0	75,183
Total	127,096	500	0	126,595

Comparison of ex-ante and ex-post realized net GHG emission reduction

Estimated ex-ante emission reduction for the monitoring period (29/06/2021- 30/09/2022)	Actual emission reduction for the monitoring period (29/06/2021- 30/09/2022)
132,424	126,595

APPENDIX 1:



Document 1: Energy Generation License of TOHMA Biomass Power Plant from the Republic of Turkey Energy Market Regulatory Authority (EPDK)

APPENDIX 2:

T.C.	
ENERJİ VE TABİİ KAYNAKLAR	
BAKANLIĞI	
KİSMİ KABUL TUTANAĞI	
TOHMA BİYOGAZ	
ENERJİ LİMİTED ŞİRKETİ	
TOHMA BES	
1 x (1,605 MW_m / 1,560 MW_e)	
GMG1 No'lu (Gaz Motoru-Generatör Grubu)	
LİSANS TARİHİ VE NUMARASI	: 10/06/2021 – EÜ/10260-7/04952
LİSANS TAKİ TESİS İLİ	: MALATYA
LİSANS TAKİ TESİS TİPİ	: YENİLENEBİLİR ENERJİ
LİSANS TOPLAM KURULU GÜCÜ	: 12,320 MW _m / 12,000 MW _e
LİSANS ÜNİTE SAYISI VE KURULU GÜCÜ	: (8,655+2,060+1,605 MW _m) / (8,440+2,000+1,560 MW _e)
KABULE ESAS TOPLAM KURULU GÜÇ	: 1 adet 1,605 MW _m / 1,560 MW _e
KABULE ESAS ÜNİTE SAYISI VE KURULU GÜCÜ	: 1 adet x (1,605 MW _m / 1,560 MW _e)
BU KABUL İLE ULAŞILAN İŞLETME GÜCÜ	: 1,605 MW _m / 1,560 MW _e
PROJE ONAY TARİH VE SAYISI	: 24.06.2021 - 65737
29.06.2021	

Document 2: Energy Generation License of the TOHMA Biomass Power Plant from the Republic of Turkey
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